

ALTA SCUOLA POLITECNICA

XXII CYCLE (2026–2027)

BOOK OF PROJECTS

FOR ASP STUDENTS

Contents

1	<i>PULSE</i>	7
2	<i>Navisio AI</i>	11
3	<i>AI & Semiconductor in Space</i>	13
4	<i>P.R.E.M.I.A. 57 (Point Cloud Regression Estimator for Molding Injection Analysis)</i>	17
5	<i>AI-Enabled Operational Intelligence in the Third Sector: Designing the Minimum Architectural Conditions for AI Adoption in Multi-Database Socio-Health Ecosystems</i>	21
6	<i>AI-assisted Structural Digital Twins for Risk-informed Retrofit Planning of Public Building Portfolios</i>	25
7	<i>Rethinking Agrivoltaics as a Climate-Resilient System</i>	29
8	<i>CO-HOUSING SENIOR</i>	33
9	<i>Staffing&Scheduling</i>	35

- 10 *EASE - Enhanced Automotive Simple Experience* 39
- 11 *Mobile Robotics - sviluppo soluzioni innovative* 43
- 12 *A Building Digital Infrastructure: rethinking interior retrofit through modularity and digital twins.* 47
- 13 *RiGOMMA: Circular Economy for Safety Rubber Recycling* 51
- 14 *POLAR-IF Polar Italian-French Antarctic Research Summer Camp* 55
- 15 *"AdhereAI": AI-Powered Patient Adherence Engine for Pharmacological Obesity Therapies (GLP-1, GIP)* 59
- 16 *An Autonomous Airship Docking Platform for Small Unmanned Multi-Copters* 63
- 17 *SCOUT – Smart Classification of Opportunities Using Tourism intelligence* 67
- 18 *Smart Lighting in Museums: Development of an Innovative Light Sensing and Adaptive Control System for Palazzo Ducale in Mantua* 71
- 19 *DATA CENTERS AS INCUBATORS OF INNOVATION ON SLOW, RESILIENT TERRITORIES* 75
- 20 *How Can We Shape Waves? Learning from Nature to Control Sound in Water* 79
- 21 *Production Process Optimisation with AI* 83

- 22 *Powering IoT with Indoor Light: Integrated Energy Harvesting and Storage Device for Autonomous Sensors* 87
- 23 *SWARM-SHELL Swarm Robotics for Additive Manufacturing of Shell Structures* 91
- 24 *Il Motore Digitale dell'Organizzazione del Futuro* 95
- 25 *Semi professional high-speed-cooking oven* 99
- 26 *RANGE EXTENDER FOR SMALL UAV SYSTEMS* 103
- 27 *Scaling Hybrid storage for Innovative Future energy Technologies (SHIFT)* 107
- 28 *Rural Proximities. Designing Living Systems for Food, Energy and Multi-species Coexistence* 111
- 29 *Modello di rigenerazione dei suoli urbani e peri-urbani* 115

1

PULSE

Contact Information

Project type: Industry & Innovation oriented

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Overall Project Objectives

The PULSE project aims to strengthen AIRC's sense of belonging and foster a culture that is open to change through a combination of active listening, internal communication, employee engagement, and change-related training. Its objectives are (i) to understand employee needs through surveys, focus groups and continuous feedback; (ii) to communicate clearly and consistently via multichannel internal campaigns; (iii) to involve people directly through workshops, internal communities and change ambassadors; and to support mindset evolution through training on agility, resilience and inclusive leadership.

General Project Description

PULSE is an integrated program designed to strengthen the sense of belonging within AIRC and support the organization in embracing a culture open to change. It emerges from the need to better understand employees' expectations, reduce cultural resistance, and improve collaboration across teams. The project responds to organizational challenges such as limited participation in corporate initiatives, uneven adoption of change, and the need for more continuous and structured listening. Its motivation is to create an environment where people feel heard, engaged, and equipped to navigate transformation. Stakeholders and Their Roles The project involves multiple organizational actors. The Project Owner is the Head of Organizational Development, responsible for coordinating activities and ensuring

alignment with organizational goals. The COO acts as Sponsor, providing strategic endorsement and oversight. Operational members include HR, Organizational Development, and IT, contributing respectively to listening tools, communication formats, and digital platforms. Stakeholders include the Executive Team, the Management Team, and the entire staff, who participate in surveys, workshops, communities, and training activities. This broad involvement ensures that the project reflects real needs and drives shared ownership of cultural evolution.

Expected Timeline and Key Phases Although structured over a full implementation cycle, the project follows a sequence of phases focused on learning, engagement, and consolidation.

- Analysis and Planning: identification of employee needs and definition of project goals and streams.
- Design: refinement of the plan, validation with leadership, and preparation of tools for communication, listening, and engagement.
- Implementation: activation of initiatives such as surveys, campaigns, workshops, communities, and training on agile mindset, resilience, and inclusive leadership.
- Launch: amplification of communication efforts and involvement actions through multichannel outreach.
- Monitoring and Synthesis: evaluation of participation rates, communication outputs, and engagement levels, followed by a structured synthesis of results, benefits, and best practices for leadership.

Through this structured flow, PULSE aims to embed listening, participation, and cultural openness into AIRC's everyday working environment, ensuring sustained organizational development.

Expected Results

PULSE is expected to deliver a set of concrete outcomes that directly address the organization's need for stronger belonging, clearer communication, and greater openness to change. The program will generate actionable insights through internal surveys, focus groups, and continuous feedback mechanisms, enabling leadership to understand employees' needs more accurately. It will also produce structured internal communication campaigns and engagement initiatives—such as workshops, internal communities, and change ambassadors—that increase transparency, alignment, and participation across the organization. These results address key challenges: low participation, cultural resistance, and the need for improved collaboration. By increasing engagement, strengthening AIRC's internal climate, and reducing barriers to change, PULSE enhances organizational effectiveness and contributes to higher retention, productivity, and innovation capacity.

Innovation and Multidisciplinarity

PULSE is innovative because it unifies listening, communication, engagement and change-enablement into a coordinated program rather than treating them as separate initiatives. This integration creates a dynamic, continuous loop of feedback, communication and behavioural support that strengthens belonging and accelerates cultural openness to change. The combination of surveys, focus groups, multichannel campaigns, workshops, internal communities, ambassadors and tailored training represents a comprehensive and modern approach to organizational culture development. The project requires the collaboration of multiple disciplines: HR contributes expertise in people processes and training; Organizational Development shapes engagement formats and change methodologies; IT supports digital tools for surveys, intranet communication, and community platforms; internal communication ensures message clarity and coherence; leadership teams provide direction and sponsorship. These perspectives complement each other by aligning data-driven insights, communication channels, behavioural levers and digital enablers ensuring that cultural transformation is broad and sustainable.

2

Navisio AI

Contact Information

Project type: Business oriented (entrepreneurship & high tech start-ups)

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Overall Project Objectives

Navisio AI is an AI assistant for the insurance industry sold on the market. It is a SaaS software that reads and analyzes policies, information sets, and quotes, extracts key points, and compares coverage, limits, and exclusions, reducing time and errors. Clear and traceable outputs for brokers and agents.

General Project Description

We have already launched Navisio AI, a vertical artificial intelligence assistant for the insurance industry, and it is already on the market with paying customers. The solution is a SaaS software designed to support and enhance the daily work of brokers, agents, and claims offices: it reads and interprets complex documents (policies, conditions, information sets, appendices, estimates), extracts relevant information, and transforms it into actionable responses, drastically reducing processing times and interpretation errors. Navisio AI is not a generic chatbot: it is built for a compliance-critical context, where accuracy and reliability matter. It combines a public insurance knowledge base and the organization's private library (internal documents and materials) to make consultation, research, and product comparisons much faster, more traceable, and consistent. In short: Navisio AI automates and accelerates the document-related activities that currently slow down brokers' operations, freeing up time and capacity for consulting, sales, and effective client management.

Expected Results

allow those working on the project to help us improve the AI engine and the software

Innovation and Multidisciplinarity

Knowledge of full stack development (software engineer), database and cloud knowledge, and minimal knowledge of artificial intelligence

3

AI & Semiconductor in Space

Contact Information

Main contact: Jonathan Bizzi; Daidalos srl; via Cesare Pavese 24
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Project type: Business oriented (entrepreneurship & high tech start-ups)

Overall Project Objectives

The project aims to define a structured market entry strategy for Daidalos within the Space Economy sector, focusing on semiconductor and AI technologies for space applications. The objective is to map potential customers, key stakeholders, regulatory requirements, and technical standards relevant to AI computing technologies in space (satellites, payloads, edge AI, robotics, Earth observation systems). The team of Management Engineering students with a Space Economy focus will analyze the industrial ecosystem, space qualification requirements, ESA/EU/FAA regulatory frameworks, and business opportunities at Italian, European, and global levels, supporting Daidalos in positioning its AI hardware solutions within the space market.

General Project Description

The project aims to provide Daidalos with a specialized market analysis to support its entry into the Space Economy sector, with a particular focus on AI and semiconductor technologies for space applications. Context and Motivation Daidalos is a deep-tech startup developing a high energy-efficiency AI accelerator. The space market increasingly requires robust, reliable, and energy-efficient computing technologies for satellite payloads, onboard edge AI systems, robotic controllers, and complex missions such as Earth observation, telecommunications, and autonomous space systems. However, the

Space Economy presents significant regulatory and technical barriers, including stringent qualification standards, certification processes, and industry-specific compliance requirements. Stakeholders and Roles Students will conduct:

- Market analysis across space segments (LEO, GEO, exploration, Government and Commercial markets), identifying target customers and global competitors;
- Stakeholder mapping including ESA, system integrators, and prime contractors (e.g., Airbus Defence & Space, Thales Alenia Space, Lockheed Martin, etc.);
- Regulatory and standards analysis covering space requirements (ECSS), avionics standards (DO-160/DO-178), export regulations (ITAR/EAR), and NASA/ESA qualification frameworks;
- Identification of technical requirements for semiconductors and AI hardware in space, including radiation tolerance, power constraints, thermal management, and reliability testing.

Timeline and Key Phases

Project setup and stakeholder identification	(2 weeks)
Desk research and expert interviews	(4 weeks)
Data synthesis and competitive benchmarking	(3 weeks)
Final report and strategic presentation	(3 weeks)

Expected Results

- Comprehensive market analysis of AI and semiconductor segments within the space industry;
- Identification and mapping of key stakeholders (ESA, prime contractors, EU/Italian space clusters);
- Structured report on regulatory frameworks and technical qualification standards for space-grade technologies;
- Competitive benchmark of AI computing solutions currently deployed or under development for orbital environments;
- Strategic recommendations to position Daidalos within the space value chain, including engagement pathways with ESA and global partners.

Innovation and Multidisciplinarity

Innovation: The project integrates strategic management, space economy analysis, semiconductor technology, and AI hardware expertise. It addresses a critical knowledge gap regarding the commercialization of advanced AI computing technologies within the Italian and European space ecosystem. **Multidisciplinarity:** Combining economic analysis, market research, and advanced technical understanding of AI hardware and space qualification requirements is essential to produce actionable strategic outcomes for Daidalos and relevant stakeholders (e.g., prime contractors, ESA). **Complementarity:** The management perspective enables integration of regulatory analysis, business modeling, and stakeholder mapping with technical

constraints and performance requirements, generating structured insights to support strategic decision-making.

External Institutions and Tutors

Institutions: ESA BIC Torino; i3P – Politecnico di Torino Innovation Company

Principal tutors: Jonathan Bizzi - jonathan.bizzi@daidalos.it - Daidalos srl; ESA BIC Torino/ I3P - Ludovico Del Carretto - delcarretto@i3p.it; ESA BIC Torino/ I3P - Leo Italiano - l.italiano@esabicturin.it

4

P.R.E.M.I.A. 57 (Point Cloud Regression Estimator for Molding Injection Analysis)

Contact Information

Project type: Industry & Innovation oriented

Main contact: Stefano Patisso, CTO MeC srl, Via Terni 2, 10149, Torino (TO)

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Overall Project Objectives

The project aims to develop an AI-driven system for calculating the cost of plastic injection molds by analyzing 3D models. The primary goal is to standardize the cost analysis methodology, ensuring repeatability and drastically reducing manual errors and processing times. By leveraging neural networks, the system will automatically identify key physical and geometric cost drivers—such as undercuts—to predict mold dimensions and machining hours based on historical data. This project represents a shift from subjective estimation to data-driven manufacturing intelligence. Innovation lies in the integration of geometric deep learning with industrial cost analysis. Success requires a multidisciplinary synergy between data scientists, who design the neural networks, and cost analysts, who provide the domain expertise needed to map physical features to actual manufacturing costs. The main beneficiaries include:

- Cost Analysts and Sales Teams: For faster, more accurate estimates.
- Team Leaders and Project Managers: To capitalize on company know-how and optimize resource planning.
- In general for our Company to gain a competitive edge through standardized and scalable analysis.

General Project Description

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- **Cost Analysts and Sales Teams:** For faster, more accurate estimates.
- **Team Leaders and Project Managers:** To capitalize on company know-how and optimize resource planning. In general for our Company to gain a competitive edge through standardized and scalable analysis.

Stakeholders and Roles:

- **CTO:** provide the industrial context and requirements.
- **Head of Chemical Costing Team:** ensure compliance with internal cost estimate process and quality of expected results.
- **Tooling Subject Matter Expert:** provide the expertise and “know-how” to train and validate the AI models.
- **ICT Team:** support development and implementation of the solution.

Expected Timeline and Phases:

1. **Data Collection & Cleaning:** Consolidating existing 3D models and cost breakdowns (currently work in progress by MeC Team). Expected duration: to be completed by March 2026.
2. **AI Model Development:** Designing neural networks for feature extraction and cost parameter estimation. This phase has already started, with an expected duration of 9 months.
3. **Testing & Validation:** Comparing AI-generated estimates against historical manual data to ensure accuracy and repeatability. Expected duration: 6 months (to be managed in parallel with phase 2).
4. **Deployment:** Integrating the system into the standard workflow to reduce manual effort. Expected duration: 3 months.

Expected Results

The project aims to deliver an AI-driven cost estimation system that transforms how MeC srl evaluates plastic injection molds. The primary deliverables include:

- **Integrated 3D-Feature Database:** a centralized repository housing historical 3D models and their corre-

sponding cost breakdowns to capitalize on existing manufacturing know-how. • Automated Feature Recognition Tool: module capable of identifying critical cost drivers, such as undercuts, directly from geometric data. • Neural Network Estimator: a predictive model that outputs key parameters like mold dimensions and machining hours based on identified features. This solution directly tackles the inefficiencies of manual estimation. The automation of physical and geometric analysis will significantly reduce lead times for quotes and minimize human error, providing a scalable, data-driven approach to complex manufacturing challenges. By leveraging historical data and geometric analysis, the project tackles also the challenge to capitalize on manufacturing know-how, ensuring that the costing methodology, know-how and years of expertise are digitally preserved and consistently applied.

Innovation and Multidisciplinarity

This project shifts mold cost estimation from manual, experience-based estimate to automated workflow. Unlike traditional methods that rely on parametric spreadsheets, this solution uses neural networks to analyze 3D models directly, identifying complex geometric cost drivers like undercuts that could be missed or miscalculated by human operators. By digitizing and capitalizing manufacturing know-how, it ensures high repeatability and a drastic reduction in analysis time. The project integrates three core disciplines: • Cost Analysis: to define the physical constraints and manufacturing features which represent cost driver of plastic injection molding. • Data Science: to build the database, develop and train neural networks capable of predicting key variables such as mold dimensions and machining hours. • Cost Modelling and Management: to standardize cost methodologies and provide consistent results. These disciplines are complementary to each other and their integration is critical: Cost analysis provides the technical solid expertise, Data Science transforms this into predictive power and Cost Management ensures the outputs are aligned with business cost-efficiency goals.

External Institutions and Tutors

Institutions: MeC srl

Principal tutors: Stefano Patisso, CTO MeC srl, Via Terni 2, 10149, Torino (TO)

Additional tutors: Giulia Beghini, Head of Chemical Cost Analysis, giulia.beghini@mecsrl.net, MeC srl

5

AI-Enabled Operational Intelligence in the Third Sector: Designing the Minimum Architectural Conditions for AI Adoption in Multi-Database Socio-Health Ecosystems

Contact Information

Project type: Social oriented (social impact and engagement)

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Overall Project Objectives

The project aims to define and validate a replicable model — the Minimum Viable Data Infrastructure (MVDI), defined as the minimum set of data models, governance rules, and technical components needed to make an information ecosystem queryable without invasive changes to source systems — enabling socio-health Third Sector organizations to govern their data and make it queryable with AI. The case study is Cooperativa Frassati (Turin): 809 workers, 93 active projects, €34.7M revenue, 7 heterogeneous, non-integrated information systems. The cooperative is part of Consorzio Merak (9 organizations), providing an immediate and practical replication context. Multidisciplinarity is essential: the problem is not technological but architectural, organizational, and economic. Without an integrated approach, AI risks amplifying existing inconsistencies rather than generating value. Beneficiaries: Third Sector organizations (1 direct + 9 in Consorzio Merak), social care workers, end users of services, and — through the replicable model — the entire Italian welfare ecosystem

General Project Description

CONTEXT AND MOTIVATION In the socio-health Third Sector, digitalization is widespread but not strategic: multiple systems, non-integrated, lacking a shared data model. Every managerial analysis requires manual extraction and reconciliation from multiple sources, taking hours. AI cannot be applied to an uncertified data foundation — without a minimum level of semantic consistency, it risks producing apparently reliable but decisionally misleading insights. The project does not aim at replacing existing systems or achieving full system integration, but at designing a lightweight data layer — the MVDI — that makes data reliably queryable without invasive changes to source systems.

STAKEHOLDERS AND ROLES - Cooperativa Frassati: case study, real-world context provider, corporate tutor - Consorzio Merak (9 organizations): replication ecosystem - ASP team (5-6 multidisciplinary students): research, modelling, prototype development - Academic tutors PoliMI / PoliTO: scientific supervision

PHASES AND TIMELINE (April 2026 – May 2027) Phase 1 — Data Discovery (Apr–Jun 2026): system census, data structure analysis, quality assessment. Output: Data Inventory + Requirements & Feasibility Assessment. Phase 2 — Architectural Modelling (Jul–Sep 2026): minimal integrated data model, semantic layer, governance rules. Output: MVDI — Minimum Viable Data Infrastructure. Phase 3 — AI Decision Support / Demonstrator (Oct 2026–Feb 2027): controlled AI interface, source traceability, uncertainty management, audit trail. Purpose: reliable querying, not prediction. Output: Working prototype on certified perimeter. Phase 4 — Impact & Replicability (Mar–May 2027): reduction of manual reporting time, decision transparency, scalability roadmap. Output: Replicable guidelines for the Third Sector.

Expected Results

MAIN DELIVERABLES 1. Data Inventory of the cooperative's information systems 2. Requirements & Feasibility Assessment: mapping of managerial questions to available data, reliability, and AI applicability 3. Minimum Viable Data Infrastructure (MVDI): data model, semantic layer, governance rules 4. AI querying prototype on certified perimeter (use cases: Workforce & Scheduling; Cost-centre Sustainability) 5. Replicable guidelines for Third Sector organizations

EXPECTED IMPACT - Reduction of manual reporting time (estimated: from hours to minutes for recurring analyses) - Improved transparency and traceability of operational decisions - Model immediately extendable to the 9 organizations of Consorzio Merak

(HR systems already shared on Zucchetti consortium platform) -
Replicable framework for the entire Italian non-profit socio-health
sector

Innovation and Multidisciplinarity

INNOVATION The project addresses the real bottleneck of AI adoption in the Third Sector: not the technology, but the information architecture. It introduces a framework — the MVDI — designed for organizations with incrementally grown systems and limited resources for structural integration. It does not propose an isolated technological solution, but rather a paradigm shift: defining the minimum conditions that make AI applicable and sustainable. Replicability is tangible: Consorzio Merak provides an ecosystem of 9 organizations with already partially shared administrative infrastructure. **MULTIDISCIPLINARITY** The problem simultaneously requires: - Computer Science Engineering / AI: data architecture, semantic layer, applied Large Language Models (LLMs) - Management Engineering: data governance, accountability, decision processes, welfare economics, organizational impact measurement None of these disciplines alone is sufficient. Their integration is not accessory but constitutive: a mono-disciplinary team cannot address the real complexity of this case study

External Institutions and Tutors

Institutions: Consorzio Merak Società Cooperativa Sociale

Principal tutors: Luca Bertone - l.bertone@coopfrassati.com - Consorzio Merak Società Cooperativa Sociale

AI-assisted Structural Digital Twins for Risk-informed Retrofit Planning of Public Building Portfolios

Contact Information

Project type: Industry & Innovation oriented

Main contact: Giuseppe Carlo MARANO

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Overall Project Objectives

This project develops a Generative AI "Design-to-Assembly Copilot" for long-span steel structures (roofs, bridges) built by Cimolai on global sites. It generates and compares structural alternatives—including segmentation, connections, fabrication, and erection—while respecting site constraints (remote location, transport, lifting capacity, weather). Objectives: 1) Encode a "jobsite profile" and constraint library as hard constraints for all designs; 2) Combine equation-based mechanics with physics-informed surrogate models for rapid, consistent feasibility screening; 3) Integrate a governed knowledge base with Retrieval-Augmented Generation for traceable, sourced outputs; 4) Deliver a multi-objective workflow balancing performance, constructability, cost/CO₂, and complexity, validated on a real (sanitized) case.

General Project Description

Context and Motivation Large-scale steel structures (long-span roofs, signature bridges) are increasingly built in remote or logistically complex sites, where constructability constraints—transport routes, available cranes, local welding/bolting capabilities, weather windows—heavily influence design outcomes. Current workflows address these constraints late, leading to costly redesigns or unfeasible

solutions. This project develops a Generative AI "Design-to-Assembly Copilot" that integrates site-specific constraints from the earliest conceptual phase, enabling optimal trade-offs between structural performance, cost, CO₂, and constructability. The Copilot generates and compares structural alternatives including segmentation, connections, fabrication plans, and erection strategies, all while respecting hard site constraints.

Stakeholders and Roles

Cimolai (Industrial Partner): Global leader in steel construction. Provides real-world pilot cases, site/construction expertise, and validates outputs; commits to a €15,000 contribution to ASP.

Politecnico di Torino & Politecnico di Milano (Academic Tutors): Co-supervise the 7-8 student team, bringing complementary expertise in structural mechanics, AI/data science, and design methods.

ASP Student Team: Develops the Copilot prototype, integrating constraint encoding, physics-informed surrogate models, governed knowledge base with Retrieval-Augmented Generation, and multi-objective optimization.

ASP Giunta: Oversees project alignment and reviews progress.

Timeline and Key Phases (aligned to ASP reviews)

Apr–Jul 2026 (Phase 1): Kick-off, requirements definition, jobsite profile encoding, access to pilot data. *1st review: problem framing + initial pipeline + budget plan.

Jul–Dec 2026 (Phase 2): Development of hybrid screening layer (physics + surrogate models) and governed knowledge base. *2nd review: baseline model + prototype v0.

Dec 2026–Feb 2027 (Phase 3): Multi-objective optimization integration, testing on real sanitized pilot case. Mid-term review: validated prototype v1 + preliminary Pareto solutions.

Expected Results

The project will deliver a validated "Design-to-Assembly Copilot" prototype including: Encoded jobsite profile and constraint library for site-specific manufacturability/logistics/erection constraints. Hybrid structural screening model combining physics and surrogate ML for rapid feasibility checks. Governed knowledge base with Retrieval-Augmented Generation ensuring traceable, sourced outputs. Multi-objective optimization workflow producing Pareto-optimal solutions balancing performance, constructability, cost/CO₂, and schedule. Validation report on a real sanitized Cimolai case study. How They Address Stakeholder Needs For Cimolai, results enable early identification of constructability issues, reducing costly late-stage redesigns on remote global sites. The Copilot accelerates exploration of feasible alternatives, while traceable outputs support defensible decision-making. For ASP and academic partners, the project advances physics-informed AI for construction, producing

methodologies applicable to broader infrastructure challenges. Students gain unique cross-disciplinary expertise at the intersection of structural engineering and AI.

Innovation and Multidisciplinarity

Current structural design tools address constructability constraints late, after key decisions are fixed. This project pioneers a Generative AI copilot that embeds site-specific manufacturability, logistics, and erection constraints as hard requirements from the earliest conceptual phase. It uniquely combines physics-informed surrogate models for rapid screening with a governed knowledge base using Retrieval-Augmented Generation for traceable, sourced outputs—moving beyond black-box suggestions to verifiable design alternatives with multi-objective optimization. Disciplines involved and their integration

Structural Engineering: Defines mechanics-based models, performance criteria, and constructability logic. **AI/Data Science:** Develops generative algorithms, physics-informed surrogates, and RAG systems. **Civil/Construction Engineering:** Provides site logistics, erection methods, and jobsite profiling expertise. **Industrial Design/UX:** Ensures intuitive visualization of Pareto-optimal alternatives and constraint trade-offs. **Sustainability/Management:** Integrates cost/CO₂ proxies and complexity indicators.

External Institutions and Tutors

Institutions: Cimolai SpA; Masera srl

Principal tutors: dylan Bortoluzzi, Cimolai SpA

7

Rethinking Agrivoltaics as a Climate-Resilient System

Contact Information

Project type: Industry & Innovation oriented

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Overall Project Objectives

This project aims to provide MUST with actionable, science-based design and operational guidelines for advanced agrivoltaics in Italy. The primary objective is to identify the optimal PV layout that meets regulatory requirements while maximizing crop yield and improving on-site water-cycle performance. The project will deliver a quantitative decision framework—KPIs, trade-off maps, and recommended configurations—tailored to open-field rotations (forage/potato) and Piedmont vineyards under increasing drought stress. A second objective is to evaluate integrated water solutions that create business and operational value: rainwater harvesting from PV surfaces, underground storage sizing, data-driven irrigation scheduling, and water reuse for panel cleaning or targeted cooling to increase PV efficiency. Innovation lies in turning agrivoltaics into a managed water–energy–agriculture system, enabled by a multidisciplinary approach combining hydrology, agronomy, microclimate physics, energy engineering, sensing/IoT, and data analytics. Beneficiaries include MUST, farmers/vineyard managers, technology providers, and public authorities supporting compliance and territorial resilience.

General Project Description

The rapid deployment of agrivoltaics in Italy is driven by the need to reconcile renewable energy production with the preservation of agricultural land. Agrivoltaics represents a paradigm shift in land-use planning: the same land simultaneously produces food, energy,

and ecosystem services. Under the Italian advanced agrivoltaic framework, projects must demonstrate measurable benefits in crop productivity, environmental sustainability, and improvement of the agricultural water cycle. However, the coupled interactions between photovoltaic geometry and the soil–plant–atmosphere continuum remain insufficiently quantified. This project is inherently highly interdisciplinary, integrating hydrology, agronomy, environmental and atmospheric physics, energy engineering, sensor systems design, data science, and regulatory analysis. By combining modelling, controlled experimentation, and real-field monitoring, the project aims to translate scientific understanding into operational design rules for next-generation agrivoltaic systems. MUST, an Italian company that installs and manages agrivoltaic plants, seeks science-based criteria to optimize panel spacing, elevation, and orientation, to balance energy yield, crop performance, and soil water availability. The focus is on open-field rotations and vineyards in Piemonte, where increasing drought may redefine irrigation and radiation management strategies. Agrivoltaics is thus treated not only as shading infrastructure but also as an active microclimate and water-management asset, including rainwater harvesting from PV surfaces, underground storage, optimized irrigation scheduling, and reuse for panel cleaning or cooling. Stakeholders and Roles: MUST: provides technical specifications, pilot sites, and strategic objectives aligned with advanced agrivoltaic guidelines. Regulatory and local authorities: provide compliance reference and scalability perspective. Expected Timeline (24 Months) Phase 1 (M1–6): System Framing & Co-Design. Regulatory review, KPI definition, and co-design workshop with MUST and farmers. Phase 2 (M4–12): Integrated Modelling. Simulation of radiation, wind modification, and soil water balance under alternative geometrical layouts. Phase 3 (M8–16): Experimental Validation. Wind tunnel and controlled experiments to quantify microclimate and evapotranspiration responses. Phase 4 (M12–24): Field Implementation. Deployment of monitoring systems in operational fields; development of guidelines.

Expected Results

The project will deliver validated technical, operational and strategic outputs directly addressing MUST's need for regulation-compliant agrivoltaic design under water stress. 1. Quantitative design guidelines: decision framework for panel spacing, elevation, orientation and typology. Deliverable: performance maps and trade-off curves. 2. Integrated water-management model: calibrated water-balance and evapotranspiration model for agrivoltaic layouts. Deliverable: simu-

lation tool and operational KPIs aligned with advanced agrivoltaic standards. 3. Monitoring & data blueprint: scalable sensors and data workflow. Deliverable: technical specs and deployment protocol for MUST sites. 4. Water valorisation strategies: rainwater harvesting, underground storage sizing, irrigation scheduling, and water reuse. Deliverable: techno-economic assessment and implementation roadmap. 5. Final industry guidelines & policy brief: consolidated operational recommendations for design, O&M and regulatory compliance. These outputs reduce design uncertainty, lower agronomic and regulatory risk, improve water-use efficiency and enhance energy and site resilience.

Innovation and Multidisciplinarity

This project reframes agrivoltaics as a managed water–energy–agriculture system rather than passive co-location. Innovations include: (1) a water-centred co-design that uses PV surfaces for rainwater capture, storage and targeted reuse; (2) joint optimisation of energy yield, crop productivity and soil-water availability via trade-off maps and KPIs; (3) a multi-scale validation chain (wind-tunnel, controlled experiments, field monitoring) feeding coupled physics + data-driven models; (4) industry-ready toolkits for MUST (techno-economic roadmaps, monitoring blueprints). Disciplines involved & necessity of integration Hydrology/soil physics; Agronomy/crop physiology; Environmental/atmospheric physics; Energy engineering; Sensor/IoT engineering; Data science/ML; Techno-economics and policy. How they complement each other Physics and hydrology define mechanisms; sensors and agronomy provide targeted observations; data science fuses data into predictive models; engineers translate findings into deployable designs; economics and policy ensure industrial viability and compliance—yielding a coherent, actionable framework for MUST and scalable agrivoltaic adoption.

External Institutions and Tutors

Institutions: MUST Soc. Agricola Semplice, con sede in Località Boschi, 46 - 15079, Sezzadio (AL) Società finanziatrice

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8

CO-HOUSING SENIOR

Contact Information

Project type: Social oriented (social impact and engagement)

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Overall Project Objectives

Il progetto mira a definire un modello innovativo di co-housing senior per il Comune di Cumiana, rispondendo alla crescente necessità di soluzioni abitative collaborative per la popolazione anziana. In un contesto nazionale privo di linee guida strutturate, l'obiettivo è trasporre best practice internazionali sul territorio locale. Il progetto richiede un approccio multidisciplinare per integrare l'analisi sociologica dei bisogni, la fattibilità architettonica e la sostenibilità economica. I principali beneficiari sono i cittadini anziani di Cumiana e dei comuni limitrofi, che potranno accedere a servizi alla persona integrati, contrastare l'isolamento sociale e vivere in strutture progettate su misura. Il Comune beneficerà di un modello replicabile di gestione del welfare territoriale.

General Project Description

Contesto e motivazione: L'invecchiamento della popolazione richiede nuove forme di abitare che superino i modelli tradizionali (RSA o assistenza domiciliare isolata). Il co-housing senior rappresenta una soluzione di frontiera che in Italia necessita di una fase di studio e sperimentazione per essere normata e attuata con successo. Parti interessate: • Comune di Cumiana: Fornirà i dati demografici, l'accesso alla popolazione per i rilievi e il supporto logistico. • Team ASP: Svolgerà l'analisi, lo studio dei servizi e la progettazione preliminare. • Tutor Accademici e Aziendali: Guideranno lo sviluppo

metodologico e la validazione dei risultati. Fasi chiave e tempistica: Il progetto seguirà il cronoprogramma ASP (Aprile 2026 - Maggio 2027) articolandosi in: 1. Analisi Sociale: Indagine sul campo per mappare i bisogni e la predisposizione al trasferimento della popolazione target. 2. Data Reporting: Elaborazione statistica dei dati raccolti per definire il profilo dell'utente tipo. 3. Service Design: Definizione dei servizi alla persona necessari (assistenza, socialità, logistica). 4. Progettazione Architettonica: Studio di una struttura sostenibile, inclusiva e funzionale ai desiderata emersi. 5. Presentazione Finale: Consegna del modello completo entro maggio 2027.

Expected Results

Ci si attende la consegna di un Masterplan di Co-housing Senior che includa: • Un report dettagliato sulla fattibilità sociale nel territorio di Cumiana. • Un set di soluzioni funzionali e servizi integrati scalabili. • Il concept di una struttura architettonica a basso impatto ambientale (sostenibilità) che risponda alle sfide di accessibilità e benessere. Il risultato fornirà al Comune una base tecnica e scientifica per avviare politiche abitative innovative, rispondendo direttamente alla sfida del cambiamento demografico.

Innovation and Multidisciplinarity

L'innovazione risiede nel colmare il vuoto normativo e metodologico italiano attraverso un approccio data-driven applicato al sociale. Multidisciplinarietà necessaria: • Architettura e Urbanistica: Per progettare spazi che favoriscano la coabitazione e l'efficienza energetica. • Ingegneria Gestionale/Sociale: Per l'analisi dei dati e la modellazione dei servizi. • Design dei Servizi: Per definire l'interazione tra utenti e strutture. L'integrazione di queste prospettive è fondamentale: una struttura eccellente dal punto di vista architettonico fallirebbe senza un'analisi sociale che ne garantisca l'accettazione e la sostenibilità gestionale nel tempo.

External Institutions and Tutors

Institutions: COMUNE DI CUMIANA

Principal tutors: LAURA MORELLI - assessoremorelli.comune@gmail.com
- COMUNE DI CUMIANA

Additional tutors: ANNAROSA CARELLO - annarosacarello@gmail.com
- COMUNE DI CUMIANA - SILVIO MANNA - silvio.manna@syspro.it
- COMUNE DI CUMIANA

9

Staffing & Scheduling

Contact Information

Main contact: Chiara Ceriani

Email: Chiara.ceriani@luxottica.com

Project type: Business oriented (entrepreneurship & high tech start-ups)

Overall Project Objectives

The project aims to optimize workforce management across 800 EMEA retail stores by developing an integrated ecosystem: a predictive forecasting model for ideal staffing and an automated scheduling tool considering top line KPIs (such as sales, traffic, conversion and productivity target) and exogenous factors. The main objective is to establish a closed-loop system that aligns labor supply with seasonality-driven demand, balancing cost-efficiency (productivity KPI) with high-quality service. Innovation: The innovation lies in the synergy between foresight and execution. By feeding forecasting outputs directly into an intelligent scheduling tool, the project transitions retail operations from manual, reactive tasks to an autonomous, data-driven workflow that dynamically manages complexity and local store constraints. Multidisciplinarity: This challenge demands a multidisciplinary approach: Data Science and Operations Research for predictive and algorithmic accuracy; Software Engineering for tool scalability; and Human-Centric Design to ensure the interface empowers store managers rather than merely replacing their autonomy.

General Project Description

Context & Motivation: Optimizing workforce management across 800 EMEA stores requires a dual focus: long-term financial stability and daily operational agility considering EMEA markets complexity. Indeed, with EMEA Region, there are a few markets which are

different in terms of velocity, structure, culture and sometimes IT infrastructure. The project aims to transition from current scheduling tool to a data-driven model that aligns FTE (Full-Time Equivalent) staffing with top-line financial forecasts, ensuring labor costs are both sustainable and effective. Stakeholders & Roles: • Operations & Field Teams: steering the forecasting tool to ensure it reflects on-ground realities. • Finance: Providing fiscal guardrails and ensuring budget alignment. • HR: Managing recruitment cycles based on projected staffing gaps and seasonality. • Area & Store Managers (AM/SM): primary users of the scheduling tool for daily shift optimization. Timeline & Key Phases: 1. Q3 – Strategy & Development: Finalizing the forecasting engine and staffing algorithm. 2. Q4 – Budgeting & Launch: Deploy the FTE forecasting tool for the new fiscal year budget cycle. 3. Q1 – Implementation: Rolling out the scheduling tool to AMs and SMs across EMEA. Ongoing – Re-run Cycles: use forecasting tool for continuous alignment of staffing levels with updated top-line forecasts to maintain operational efficiency.

Expected Results

The project will deliver a comprehensive Workforce Optimization Suite, comprising two core modules: a Predictive FTE Forecasting Engine to be used by home office population to budget workforce and align it to top line goals and an Automated Scheduling Interface to be used in store by SM/AM. Key deliverables include: - FORECASTING: a centralized dashboard for EMEA-wide staffing visibility to track FTE allocation at store level with a dynamic interface to slice and dice the data (data visualization layer). Moreover, the tool need to guarantee at local level to be independent to set up the main criteria based on store clusters in order to set up for yearly budget and re-run cycles based on the main drivers. - SCHEDULING: standardized tool based on main business drivers and HR rules to allow SM with a user friendly interface to know the perfect match between efficiency on shop floor and achievement of sales goals.

Innovation and Multidisciplinarity

Multidisciplinarity & Integration: Achieving this level of complexity requires the convergence of three core disciplines: 1. Data Science & Operations Research: Necessary to build the algorithmic engine for forecasting and constraint-based scheduling. 2. Corporate Finance: Essential to align labor investments with fiscal targets and ROI requirements. 3. Organizational Psychology & HR: crucial to translate mathematical shifts into ergonomic, fair, and legally com-

pliant schedules. Complementarity: these perspectives complement each other by balancing efficiency with human-centricity. While Data Science provides the "how-to" for optimization, Finance sets the "limits" of feasibility, and HR ensures "sustainability" for the workforce. Integrating these fields prevents a purely algorithmic approach from ignoring the human element of retail, ensuring the tool is both technically superior and operationally adopt

External Institutions and Tutors

Institutions: Not needed

Principal tutors: Not needed

Additional tutors: Not needed

EASE - Enhanced Automotive Simple Experience

Contact Information

Project type: Industry & Innovation oriented

Main contact: Anita Fiorentino - Stellantis Europe

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Overall Project Objectives

The project aims to rethink how users interact with onboard technologies by making vehicle functions more intuitive, accessible and immediately understandable. It addresses the gap between increasingly complex digital and assistance systems and users' expectations for simple and seamless experiences. The objective is to develop conceptual, human centred solutions that reduce cognitive load, improve clarity and increase user acceptance through multimodal interaction, guidance and gamification. Innovation stems from exploring interaction models inspired not only by automotive but also by gaming, wearables and smart home ecosystems, which set higher standards for usability and engagement. Multidisciplinarity is essential: UX design and human factors analyse behaviour; cognitive psychology supports understanding of perception and mental models; engineering and AI ensure feasibility; digital communication shapes coherent and engaging experiences. Primary beneficiaries include end users, who will gain more intuitive and engaging interactions, and Stellantis, which can integrate these insights into future user-centred mobility solutions.

General Project Description

The project addresses the growing gap between the increasing complexity of onboard digital, connected and driver assistance systems and users' expectations for simple, intuitive and seamless interactions. Many drivers struggle to understand available features or navi-

gate interfaces perceived as fragmented and cognitively demanding. The motivation is to rethink how people discover, learn and interact with vehicle functions by exploring conceptual, human centred solutions that reduce cognitive load and enhance clarity, acceptance and perceived value. The main stakeholder of the project is Stellantis, which aims to increase the customer appeal and perceived value of its vehicles through more intuitive and meaningful interaction models. Depending on the project evolution, different internal functions may contribute by providing insights on user needs and interaction challenges. Additional stakeholders may include consumer associations and organisations involved in collecting customer feedback on technological usability and expectations. Their perspectives, gathered through literature reviews, surveys or targeted consultations, can help integrate broader user viewpoints into the project framework. The project follows the 17 month ASP timeline. In addition to the official milestones, monthly alignment meetings will be held between students, academic tutors and Stellantis to monitor progress and refine research directions. A first milestone is planned for July 2026, when the team will consolidate insights on what users expect from onboard technologies and what solutions could realistically be offered. A second milestone is scheduled for October 2026, when the most promising concepts will be selected and developed into a proof of concept not integrated into a vehicle, enabling preliminary evaluations and user centred analyses. The final assessment phase is expected for February 2027, including the evaluation of the PoC and the consolidation of recommendations ahead of the final submission in May 2027.

Expected Results

The project is expected to deliver three main results. 1) A detailed analysis of user expectations, current pain points and desired future experiences, identifying what today's users find valuable, what they consider difficult or confusing, and what digitally native students would expect from a modern, intuitive and seamless interaction with onboard technologies. 2) An analysis of technological opportunities—within and beyond automotive—leveraging the fresh, connected perspective of young students. This comparison between user needs and available technologies will support the selection of the most promising interaction concept. 3) A proof of concept (PoC) representing the selected solution, enabling early evaluations of usability, perceived complexity, emotional response and overall user acceptance. These results address Stellantis' needs by providing clear insights into how users understand and experience onboard technolo-

gies and by identifying interaction strategies that simplify interfaces, enhance feature discoverability and strengthen perceived value. The outcomes will support the development of more intuitive, consistent and user centred mobility experiences.

Innovation and Multidisciplinarity

The project addresses a key gap in current automotive experiences: despite the availability of advanced digital and assistance functions, users rarely consult static manuals and often misunderstand or underuse available features. Existing interfaces do not meet with user'expectations . EASE proposes a shift toward conceptual, multimodal and personalised models inspired by gaming, wearables, smart home ecosystems and other non automotive domains, aiming to reduce cognitive load, increase clarity and enhance user acceptance. Multidisciplinarity is essential. UX design and human factors analyze behavior, clarity and cognitive effort; engineering and AI provide feasibility and system logic; communication and digital design create coherent and engaging interaction narratives. These perspectives complement each other, enabling solutions that are both visionary and realistic, ensuring that conceptual models, interaction guidelines and the PoC reflect a comprehensive and user centered viewpoint, with additional inputs, if needed, coming to Stellantis' internal functions.

External Institutions and Tutors

Institutions: Centro ricerche Fiat and Stellantis Europe

Principal tutors: Anita Fiorentino – anita.fiorentino@stellantis.com - +39 3356908820; Maurizio Luppino - maurizio.luppino@stellantis.com

Additional tutors: to be identified, if needed within Stellantis

Mobile Robotics - sviluppo soluzioni innovative

Contact Information

Project type: Business oriented (entrepreneurship & high tech start-ups)

Main contact: Nicola Dello Stritto. Pramac. loc il piano, colle val d'elsa, Siena a

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Overall Project Objectives

The project aims to identify and evaluate complementary solutions that can expand the mobile robotics portfolio of PRAMAC and enhance the value of existing systems. It combines market analysis, robotics engineering, and industrial logistics to generate innovation opportunities. The outcomes will benefit the company's innovation roadmap and provide value to industrial customers seeking integrated automation solutions.

General Project Description

The project focuses on exploring and defining complementary solutions to expand the current portfolio of mobile robotics products developed by PRAMAC. As the market for mobile robotics and automated material handling continues to grow, customers increasingly require integrated and flexible solutions rather than standalone robotic systems. The motivation of this project is therefore to identify and evaluate additional products, technologies, or system components that can complement the existing autonomous mobile robots and improve the overall value proposition for industrial customers. The project will involve several internal stakeholders. The Product Management function will lead the project, defining the strategic direction, identifying market opportunities, and coordinating the overall activities. Application Engineering will contribute technical expertise and customer insights related to system integration and use

cases. The R&D department will provide technical validation of potential solutions and assess feasibility. In addition, selected external stakeholders such as technology partners, suppliers, and potential customers may be involved to gather feedback and market perspectives. The project is expected to run over a period of approximately 4–6 months. The first phase will focus on market and technology analysis, including benchmarking of complementary solutions. The second phase will involve the identification and evaluation of potential concepts or product opportunities. In the final phase, the most promising options will be consolidated into a structured proposal including business rationale, potential applications, and high-level development recommendations

Expected Results

The project is expected to deliver a structured analysis of potential complementary solutions that could extend current mobile robotics portfolio of PRAMAC. The main objective is to identify opportunities that can enhance the value proposition of existing robotic systems by addressing additional customer needs in automated material handling and intralogistics. The key deliverables will include several outputs. First, a market and technology overview identifying relevant trends, competing solutions, and emerging technologies related to mobile robotics ecosystems. Second, a benchmarking analysis of complementary products or systems that can integrate with autonomous mobile robots, such as software tools, accessories, or supporting equipment. Third, a structured evaluation of selected opportunities based on criteria such as market potential, strategic fit, technical feasibility, and potential value for customers. The final outcome of the project will be a prioritized shortlist of potential solutions or concepts, together with a high-level business rationale and preliminary recommendations for further development or partnership exploration.

Innovation and Multidisciplinarity

The innovative aspect of the project lies in its focus on expanding the ecosystem around mobile robotics rather than developing standalone robotic units. While many existing initiatives concentrate primarily on improving the performance of autonomous mobile robots, this project explores complementary solutions that can enhance the usability, flexibility, and integration of robotic systems within industrial environments. By identifying additional products, technologies, or services that can interact with mobile robots, the

project aims to move toward a more integrated and solution-oriented approach to intralogistics automation. This perspective is particularly relevant for companies such as PRAMAC, where the competitive advantage increasingly depends on offering complete systems that address broader operational needs rather than individual machines. The project involves multiple disciplines. Product management contributes market analysis, strategic positioning, and evaluation of customer needs. Engineering and robotics provide technical knowledge related to system architecture, hardware integration, and automation technologies.

External Institutions and Tutors

Institutions: Pramac

Principal tutors: Nicola Dello Stritto, nicola.dellostritto@pramac.com, Pramac.

A Building Digital Infrastructure: rethinking interior retrofit through modularity and digital twins.

Contact Information

Main contact: Agata Bandini

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Project type: Business oriented (entrepreneurship & high tech start-ups)

Overall Project Objectives

The project aims to develop a constructive digital infrastructure for interior renovations based on a modular building open system integrated with digital twin technology. The objective is to innovate the way we conceive and build interiors through Design for Manufacturing and Assembly principles, reducing demolition, waste and uncertainty in retrofit operations. The project seeks to demonstrate how standardized components can remain adaptable throughout their lifecycle through parametric design and digital monitoring tools. Innovation lies in combining constructive modularity with the power of new digital technologies such as Digital Twins, transforming interior architecture into an active infrastructure rather than a static partition, able to transform over time while transmitting information and predicting problems. The multidisciplinary approach integrates architecture, engineering, computational design and business modeling. Beneficiaries of the project include property owners, retrofit contractors and real estate operators.

General Project Description

The building sector accounts for a significant share of global CO₂ emissions, while renovation rates remain too low to meet 2050 climate targets. Retrofit processes are still largely artisanal, unpre-

dictable and difficult to scale. Innovation has focused mainly on envelopes and energy systems, leaving interior transformation largely unresolved. Yet interior demolition generates substantial waste and limits spatial adaptability over time. This project addresses retrofit from the inside, rethinking partitions as an active infrastructure. Instead of permanent masonry divisions, we propose a dry-assembled wooden system designed according to Design for Manufacturing and Assembly principles and organised in a catalogue of solutions. Parametric components are designed to be standardised but adaptable to any specific case thanks to an integrated parametric configurator. Thanks to this technology, the drawing can be sent directly to production, digitally fabricated and then ready for mounting, dismounting and reuse. The project integrates in fact a digital twin platform that accompanies the system across its lifecycle, from design until realisation. After the phase of parametric design, the construction process begins with digital surveying (laser scanning), followed by a traditional yet minimised preparation phase. The parametric configurator then allows accurate reconfiguration and optimization. CNC production and dry on-site assembly complete the installation. Over time, the digital twin supports maintenance, monitoring and future spatial transformations. Stakeholders include: 1) Industrial partners for CNC fabrication, who can benefit from the process optimization enabled by the Digital Twin; 2) Technology partners for laser scanning and IoT integration; 3) Real estate or renovation actors as potential pilot users, as well as municipalities. The timeline is structured in three phases: 1) System definition and parametric modelling; 2) Prototype development and digital twin MVP; 3) Testing, environmental assessment and market positioning.

Expected Results

The project will deliver: 1) A 1:1 prototype of a modular wall system based on DfMA principles; 2) A minimum viable digital twin platform enabling configuration and monitoring; 3) Parametric design rules adaptable to different spatial constraints; 4) A preliminary environmental comparison between demolition-based and incremental renovation scenarios (LCA); 5) A conceptual business model for market introduction. These outputs respond to the need for scalable, predictable and low-impact retrofit solutions. They support organizations seeking faster renovation processes, reduced waste, improved cost control and integration of digital lifecycle management tools. The project provides a replicable framework that bridges construction, digital technology and market feasibility

Innovation and Multidisciplinarity

The project is innovative because it shifts interior renovation from demolition-based transformation to incremental architectural adaptation. Rather than adding static elements to invariable interiors, as in traditional construction, it integrates construction logic and digital lifecycle management into a single system. Partitions therefore have the potential to become infrastructural elements combining modular fabrication, parametric configuration and digital twin monitoring. Disciplines involved include architecture (spatial system design), civil engineering (constructive feasibility and assembly logic), computer science (parametric modeling, digital twin and AI-based configuration tools), and management engineering (industrial process design and business modeling). Their integration is essential: constructive modularity requires digital precision; digital twins require spatial constraints; and both require economic viability to become scalable solutions. The project demonstrates how physical and digital systems can co-evolve within a unified retrofit strategy.

External Institutions and Tutors

Institutions: We are currently involving FederLegno, EDERA and DesignTech for enriching our project proposal.

Principal tutors: 1. Agata Bandini, agata.bandini@ladomus-design.com, Lodomus / 2. Edoardo Lo Vecchio, edoardo.lovecchio@ladomus-design.com, Lodomus

RiGOMMA: Circular Economy for Safety Rubber Recycling

Contact Information

Project type: Industry & Innovation oriented

Main contact: Loris Marchisio - Fondazione WellFARE impact - Via Pollino, 4 - Fraz, 12100 Ronchi CN

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Overall Project Objectives

The project aims to promote a regeneration process for shock-absorbing rubber tiles used to cover the flooring of public playgrounds. Once these tiles reach the end of their life cycle, they are normally destined for landfill, yet they can be transformed into a new resource. The main objective is to develop an innovative recovery and redesign process that extends the product's life cycle, reduces waste, and supports a more sustainable production model, fostering a regenerative rather than extractive economy. The innovation lies in integrating material-analysis techniques, circular design principles, and new transformation technologies capable of generating solutions that can be replicated in other sectors as well. A multidisciplinary approach is essential: engineering, ecodesign, circular-economy expertise, and environmental analysis work together to ensure effective and measurable results. Beneficiaries include public bodies involved in waste management, sustainability-oriented consumers, and the broader community, which benefits from a reduction in overall environmental impact.

General Project Description

The RiGOMMA project, promoted by the WellFARE Impact Foundation, introduces a circular-economy model dedicated to recovering worn-out shock-absorbing flooring from playgrounds in the province of Cuneo. Today, these surfaces—whose useful life ranges from 8 to 15 years—are almost always sent to landfill, generating high costs and a significant environmental impact. The flooring is generally made of SBR rubber granules derived from end-of-life tyres (ELTs), bound with polyurethane resins and coloured pigments, in compliance with UNI EN 1177 and EN 71-3 standards. Although several Italian companies produce recycled-rubber flooring, the cycle ends once the product reaches the end of its life: once removed, the tiles become special waste. RiGOMMA instead proposes a circular system in which discarded flooring is collected, shredded, separated by grain size, and transformed into secondary raw material that can be reused for new shock-absorbing surfaces or other recycled-rubber products. The project is based on the active involvement of local stakeholders. With GEM Chimica, a material analysis was carried out to identify the correct EWC code, understand its composition, and assess its recyclability. With ACSR, discussions were initiated to map current disposal practices and define a potential dedicated collection flow. An initial recycling test has already produced positive results, confirming the technical feasibility of the process. Based on these findings, the creation of an experimental laboratory has been proposed to optimise the shredding, separation, and reassembly phases, with the aim of defining a stable and replicable protocol. Once the process is consolidated, work will be needed to redefine the identity of the secondary raw material, identifying new applications and market segments through design activities, performance analyses, and economic evaluations.

Expected Results

The project aims to develop and validate a technical process that transforms shock-absorbing rubber tiles, currently classified as waste, into a regenerated material suitable for reintroduction into production. The goal is to create a stable, repeatable and regulation-compliant procedure that enables the material to obtain product status (end of waste). Through physical, mechanical and chemical analyses, the project will define the properties of the regenerated material, establishing quality standards, performance requirements and compliant application fields. Based on these characteristics, potential uses and market sectors will be identified to map opportunities and define

new products that can incorporate the regenerated material. The project also aims to demonstrate a significant reduction in environmental impact compared to landfill disposal, improving resource efficiency and reducing waste. Ultimately, it seeks to create a replicable circular-economy model applicable to similar materials, supporting the valorisation of currently low-value waste streams.

Innovation and Multidisciplinarity

Although Italy leads Europe in recycling, a structural gap remains between material recovery and actual reuse. According to the “Quattro A” Report (February 2026) published by Il Sole 24 Ore, the national recovery rate is 98%, yet the real substitution of virgin raw materials is only 0.4%. RiGOMMA addresses this market failure by targeting shock-absorbing playground surfaces—made of SBR rubber and polyurethane binders—which currently end up 100% in landfill due to the lack of technical standards and reuse protocols. The initiative acts as a disruptive solution built on three pillars: Validation of an End-of-Waste protocol, turning the material from bulky waste into a Certified Product with defined performance standards for industrial reintroduction. A validation infrastructure in Ronchi (Cuneo), creating an advanced testing lab that enables high-precision micronisation and polymer regeneration. A shift from an extractive to a regenerative model, systematically replacing virgin raw materials and reducing pressure on natural resources by extending polymer life cycles.

External Institutions and Tutors

Institutions: Gem Chimica S.R.L. - ACSR S.P.A

Principal tutors: Graziano CERUTTI - email graziano.cerutti@gemchimica.com (GEM Chimica) - Giancarlo ISAIA - email info@acsr.it (ACSR)

POLAR-IF Polar Italian-French Antarctic Research Summer Camp

Contact Information

Project type: Research oriented

Main contact: Roberto Dini
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Overall Project Objectives

The project aims to develop a preliminary design for a new Summer Camp to support seasonal activities at the Italian–French Concordia Research Station in Antarctica. Concordia is located on the Antarctic Plateau at approximately 75°06'S, 123°23'E, at an altitude of about 3,233 m above sea level, where the average atmospheric pressure is around 650 hPa and winter temperatures can drop below –80 °C. The new Summer Camp is expected to host approximately 40 people and will likely be located closer to the station's twin towers, in coordination with the planned Emergency Camp facility. The project must address key constraints related to energy efficiency, extreme climatic conditions, integration with existing station infrastructure, and the logistical limitations associated with transporting materials from the Northern to the Southern Hemisphere, including strict limits on weight and volume. The main beneficiaries will be the Concordia station management, which may obtain a structured project concept supporting future implementation, and the participating students, who will have the opportunity to tackle unique engineering and architectural challenges.

General Project Description

The project is expected to deliver a preliminary integrated design for the new Concordia Summer Camp, including architectural concepts, functional layouts, and technical guidelines adapted to the extreme Antarctic environment. The main deliverables may include a conceptual masterplan for the camp location, preliminary architectural and structural solutions, energy efficiency strategies, and proposals for integration with the existing station infrastructure. Particular attention should be given to modularity, transportability, and assembly feasibility, considering the strict logistical constraints associated with Antarctic operations, including limited cargo capacity, weight restrictions, and long supply chains between the Northern and Southern Hemispheres. The project should also provide recommendations addressing human factors and psycho-architectural aspects to improve comfort, well-being, and social interaction in isolated and extreme environments. These outcomes will support the Concordia station management and the ENEA–UTA technical teams by providing a structured design framework that may inform future feasibility studies and engineering developments, while offering students the opportunity to engage with real-world challenges in extreme-environment infrastructure design.

Expected Results

The design of the new summer camp at Concordia Station is expected to deliver a sustainable, efficient, and adaptable infrastructure capable of supporting the seasonal increase in scientific and logistical activities. The project should provide solutions that improve habitability, operational efficiency, and safety in extreme Antarctic conditions while minimizing environmental impact. Key expected outcomes include optimized spatial organization for research, accommodation, and logistics; integration of resilient construction strategies suitable for polar environments; and the use of energy-efficient systems to reduce resource consumption. Ultimately, the design should strengthen the capacity of the Italian–French program to conduct multidisciplinary research in Antarctica while ensuring flexibility for future scientific and technological developments.

Innovation and Multidisciplinarity

The project is innovative because it addresses the design of a living infrastructure in one of the most extreme environments on Earth: the Antarctic Plateau. The Concordia station presents unique en-

vironmental and operational constraints, including extremely low temperatures, low atmospheric pressure, limited logistics, and long periods of isolation. Designing a Summer Camp in this context requires solutions that go beyond conventional architectural and engineering approaches. The project involves multiple disciplines, including architecture, civil and structural engineering, energy engineering, logistics and transportation planning, and human-centered design. The integration of these perspectives is essential to address the complexity of building and operating a seasonal facility in Antarctica. Architectural and engineering design must ensure structural resilience, energy efficiency.

External Institutions and Tutors

Institutions: ENEA Agenzia Nazionale per le nuove tecnologie, l'energia e lo sviluppo

Principal tutors: PhD Riccardo Scipinotti, riccardo.scipinotti@enea.it, Responsabile del Servizio IAP (Rapporti Istituzionali, Ambiente, Promozione e Formazione) dell'Unità Tecnica Antartide (UTA) di ENEA.

15

"AdhereAI": AI-Powered Patient Adherence Engine for Pharmacological Obesity Therapies (GLP-1, GIP)

Contact Information

Project type: Business oriented (entrepreneurship & high tech start-ups)

Main contact: Alessandra Cestaro, Founder & CEO at Holifya S.r.l.;
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Overall Project Objectives

Pharmacological therapies for obesity and metabolic syndrome, such as GLP-1 and GIP receptor agonists, are transforming chronic disease management. Yet up to 50% of patients discontinue within the first year due to side effects, suboptimal dosing, high costs, and limited clinical support, reducing outcomes and healthcare sustainability. Objectives: raise adherence to 90%+ (vs. ~50% without digital support), improve weight-loss outcomes, and enhance metabolic parameters (cholesterol, glycemia, blood pressure) through reduced side effects and stronger multidisciplinary support. The project develops four workstreams: (1) AI prediction of side effects and dose optimization using biological, genetic, medical history, and behavioral data; (2) intelligent follow-up and clinical escalation; (3) behavioral interventions via CBT, nudging, and gamification; (4) a health economics model demonstrating ROI for insurers and clinics. The team combines AI/ML engineers, biomedical experts, designers, and management engineers. Beneficiaries include patients, providers, insurers, pharma partners, and the 11M+ Italians eligible for treatment but lacking adequate care infrastructure.

General Project Description

Context & Motivation: Europe's metabolic health crisis is accelerating. Over 200M European adults suffer from metabolic syndrome, with 85M eligible for pharmacological therapies such as GLP-1 and GIP receptor agonists. In Italy, penetration stands at just 0.34% - the lowest in Europe - while demand surges (+180% in endocrinology visit requests over three years) and wait times stretch to 8 months in the public system (SSN). Holifya, Italy's first virtual clinic for pharmacological obesity treatment, has served 3,000+ patients with 15% average body weight loss at 6 months, and an exclusive partnership with Novo Nordisk. Yet across the industry, therapy dropout remains the central unsolved challenge, driven by side effects, cost, and waning motivation. PROJECT STRUCTURE & TIMELINE PROPOSED: Four integrated workstreams tackle this end-to-end: WS1 — AI Dose Optimization & Side-Effect Prediction (focus: months 1–9): ML models on Holifya's dataset (3,000+ datapoints/patient) to predict adverse events and personalize titration — targeting 55–66% medication cost reduction. WS2 — Intelligent Follow-Up & Escalation (focus: months 3–10): AI agents automating symptom monitoring, behavioral tracking, and patient education, with smart escalation to clinical staff when needed. WS3 — Behavioral Retention Engine (focus: months 4–11): CBT micro-modules, nudging, and community accountability features targeting <4% monthly churn. WS4 — Health Economics & B2B Model (focus: months 2–12): Cost-effectiveness case for insurers and clinics, outcomes dashboard, and go-to-market strategy for a licensable B2B SaaS product. Stakeholders & Mentorship Holifya (proposer, data provider, product integration), including Students will be mentored by Alessandra Cestaro, Holifya Founder & CEO (ex-McKinsey Healthcare, Columbia MBA), Holifya's senior tech team engineers with experience at Microsoft, Amazon, and Apple; and supported by Key Opinion Leaders in Healthcare, including Micol Fornaroli and Piero Gancia (key Holifya investors, ex-McKinsey), among others. Novo Nordisk (pharma partner, clinical expertise) managers will also be a key player in the project.

Expected Results

We imagine the project could deliver four key integrated outputs: 1) AI GLP-1/GIP Dosing Optimization Prototype: A validated ML model predicting side-effect risk and recommending personalized titration for GLP-1/GIP therapies - targeting 55–66% medication cost reduction in medication expense while maintaining clinical results effectiveness, with a clinical validation framework. 2) Intelligent

Escalation Agent: An AI prototype automating patient follow-up across three tiers (symptom triage, titration alerts, safety flags), reducing clinician workload and catching dropout signals earlier. 3) Behavioral Adherence Module: A tested prototype of CBT micro-modules, nudging, and gamification features, with an A/B testing framework measuring retention and engagement impact. 4) Health Economics & B2B Model: A cost-effectiveness analysis proving ROI for insurers and clinics, an outcomes dashboard, and a GTM strategy for licensing the AI adherence system as a SaaS product to clinics. All deliverables address Holifya's core strategic challenge and will be directly integrated into its platform, with immediate application to its 3,000+ patient base and B2B partners (Novo Nordisk, Insurance).

Innovation and Multidisciplinarity

What Makes This Project Innovative No integrated system exists combining predictive AI, intelligent automation, and behavioral science for pharmacological obesity therapy management — designed for the European regulatory context. Holifya's proprietary dataset (3,000+ datapoints/patient with structured symptom matrices) makes this uniquely feasible, pioneering the shift from digital clinic to a licensable AI-powered clinical OS. Disciplines & Integration The adherence challenge is simultaneously clinical, computational, behavioral, economic, and experiential: Computer Science & AI/ML: predictive modeling, agent design, platform engineering Bioengineering: clinical validation, drug-response modeling Management Engineering: health economics, B2B pricing, go-to-market Interaction Design: patient UX, CBT modules, care journey architecture Automation & Control: titration feedback loops, escalation logic No single discipline can solve it. AI engineers need biomedical input for safety; designers need behavioral science for effective interventions; economists need clinical data for insurer models. The project creates deliberate interdependencies requiring genuine integration.

External Institutions and Tutors

Institutions: Holifya (+ on-call support from Novo Nordisk & McKinsey & Company primarily)

Principal tutors: Holifya: Alessandra Cestaro; alessandra.cestaro@holifya.com

Additional tutors: Holifya: Vittoria Ferla (vittoria.ferla@holifya.com) as Clinical Product Manager & Medical Affairs; Akash Arora (akash.arora@holifya.com) as Senior Software Engineer / Product Engineer.

An Autonomous Airship Docking Platform for Small Unmanned Multi-Copters

Contact Information

Project type: Industry & Innovation oriented

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Overall Project Objectives

The primary objective of this project is the development and hardware–software integration of a docking and recharging station for drones mounted on a hybrid and autonomous airship. The field of LTA (Lighter Than Air) vehicles represents a growing research area, yet it is still characterized by a limited number of consolidated operational applications. The ability of these platforms to maintain altitude for extended periods with low energy consumption makes them particularly suitable as support bases for UAV systems. The airship–drone integration enables the combination of large-scale data acquisition, performed by the LTA platform while surveying areas of interest, with high-resolution local measurements carried out by drones. This approach is applicable to several domains, including geophysical monitoring, infrastructure inspection, and precision agriculture.

General Project Description

Airships represent an innovative solution for active environmental monitoring, characterized by low environmental impact and high energy efficiency. Their main strength lies in their long flight endurance, which enables persistent and extended operations. However, due

to their geometric and dimensional characteristics, which cannot be excessively reduced, these vehicles exhibit significant inertia, resulting in limited maneuverability and agility. These limitations are complementary to the capabilities of small unmanned aerial vehicles, such as quadrotor drones, which are instead characterized by high maneuverability, precise positioning, and rapid dynamic response. The integration of these two technologies can therefore be considered a natural evolution toward the implementation of a comprehensive, autonomous, hybrid, and environmentally sustainable monitoring system, capable of exploiting the strengths of both platforms. Within the scope of the project, collaborations with entities outside the academic environment are envisaged, in particular with the airship manufacturing company Flofleet, which operates from its headquarters in Sesto San Giovanni and has its production facility in Bellusco. The project begins with a state-of-the-art review of existing drone landing stations, followed by an analysis of the hardware integration of the system on the test airship provided by the industrial partner. Should this solution not be available, the construction of a dedicated aerostatic balloon for experimental purposes is foreseen. In the initial phase, the airship is considered under hovering conditions, i.e., at zero forward velocity, in order to develop, test, and validate the control systems for drone takeoff and landing operations. Subsequently, a cruise velocity of the airship will be introduced, allowing the assessment of system feasibility, the analysis of boundary cases, and the identification of the main operational and design constraints.

Expected Results

The project will deliver a validated airship–drone integration package enabling safe, repeatable drone recovery on a hybrid airship during a single monitoring mission. Expected results include: (i) an autonomous approach-and-landing guidance, navigation and control stack based on robust relative localization (sensor fusion/vision) and defined airship–drone coordination states; (ii) a resilient low-latency communication and mission management interface linking the two vehicles; and (iv) flight-test demonstrations with performance metrics (landing success rate, time-to-recover, wind/disturbance tolerance, link robustness). These deliverables address Flofleet’s need to expand monitoring capability by combining airship persistence with drone responsiveness and close-range sensing, while reducing operational cost and risk through autonomy and standardized procedures. For the university, the project provides a concrete multidisciplinary research testbed, high-value datasets from real flight trials, and publishable advances in relative navigation, control under disturbances,

and airborne docking systems.

Innovation and Multidisciplinarity

The project is innovative because it moves beyond conventional drone docking solutions, by enabling autonomous launch and recovery onto a hybrid airship acting as an airborne “mothership,” where platform motion, relative wind, and airflow disturbances make precision landing far more complex. The novelty is not only the landing pad itself, but the full system-level integration: shared navigation references, relative localization, landing authorization, health monitoring, and fail-safe behaviors that allow airship and drones to operate as a coordinated monitoring system within the same flight mission, combining long endurance and wide-area persistence with rapid, close-range, high-resolution sensing. Achieving this requires a genuinely multidisciplinary approach, integrating aeronautical engineering (airship dynamics and flight envelope), robotics and control (robust guidance and landing under disturbances), sensor fusion (reliable relative positioning beyond GNSS), telecommunications and networking (low-latency, resilient airship–drone links and state synchronization), and mechanical design and materials (capture/retention, damping, tolerances, weight and maintainability).

External Institutions and Tutors

Institutions: FloFleet, Via Enrico Toti, 2 Sesto San Giovanni (MI), Italy, info@flofleet.com

Principal tutors: Ing. Giulia Fasoli, FloFleet, giulia.fasoli@flofleet.com

17

SCOUT – Smart Classification of Opportunities Using Tourism intelligence

Contact Information

Project type: Business oriented (entrepreneurship & high tech start-ups)

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Overall Project Objectives

Hotiday is a full-stack hospitality operator managing 800+ rooms across 120+ locations in 5 European markets, grown from €900K to €16M revenue in 24 months via an asset-light model. Growth critically depends on identifying the right properties in the right destinations - currently a manual, expertise-driven process. The objective is to develop an ML-powered decision-support system to predict destination attractiveness and evaluate acquisition opportunities: (1) predictive models identifying high-potential tourism micro-markets before they peak, using public and proprietary data; (2) a property scoring algorithm estimating revenue potential for candidate assets; (3) an actionable decision-support tool for the acquisitions team. Multidisciplinarity is essential: data scientists for ML modeling, management engineers for business logic, territorial analysts for destination assessment, and designers for tool usability. Beneficiaries include Hotiday's expansion team and the broader hospitality-tech ecosystem.

General Project Description

CONTEXT AND MOTIVATION Hotiday operates as an asset-light hospitality company: it identifies underperforming hotel assets, acquires their management, and optimizes them through AI-powered

pricing, multi-channel distribution, and operational improvements. The critical bottleneck is the upstream decision, where to expand and which properties to acquire. Today, this relies on manual scouting, market intuition, and basic historical comparisons. As Hotiday scales across Europe, this approach cannot keep pace. The company needs a systematic, data-driven methodology to identify destinations with rising demand trajectories and properties with untapped revenue potential. **STAKEHOLDERS AND ROLES** Hotiday (project proposer): provides real operational datasets from 120+ managed properties (occupancy rates, ADR, RevPAR, seasonality patterns), access to public data sources already in use (Booking/Google reviews, competitor pricing, ISTAT tourism statistics), and domain expertise through a dedicated company tutor from the acquisitions/revenue management team. **ASP student team** (7-8 members): conducts research, builds models, and develops the prototype. The ideal team combines competencies in machine learning, data engineering, tourism economics, territorial analysis, and UX/UI design. **Academic tutors**: provide methodological guidance on ML techniques, economic modeling, and research rigor. **EXPECTED TIMELINE AND KEY PHASES** Phase 1 – Discovery & Data (Apr–Jun 2026): Domain immersion at Hotiday, dataset exploration and cleaning, literature review on tourism demand forecasting and real estate scoring models. Definition of evaluation criteria and KPIs. Phase 2 – Modeling (Jul–Oct 2026): Development of two core ML models: (a) a destination attractiveness predictor using macro indicators (tourism flows, infrastructure, events, connectivity, review sentiment trends), and (b) a property scoring model combining structural features with market positioning and revenue benchmarks from Hotiday's portfolio. Phase 3 – Validation & Tool Design (Nov 2026–Feb 2027): Backtesting models against Hotiday's historical acquisition decisions. Design of a user-friendly dashboard/tool for the acquisitions team. Iteration based on real-world feedback. Phase 4 – Delivery (Mar–May 2027): Final prototype, documentation, recommendations for deployment, and presentation.

Expected Results

The project will deliver three core outputs: **DESTINATION INTELLIGENCE MODEL**: A machine learning model that ranks geographic micro-markets by predicted tourism growth potential over a 1-3 year horizon. The model will integrate public data (tourist arrivals, flight/rail connectivity, event calendars, urban development plans, online sentiment trends) with Hotiday's proprietary performance benchmarks. This directly addresses the challenge of "finding localities that will perform well in coming years." **PROPERTY SCORING**

ALGORITHM: A scoring framework that evaluates individual acquisition candidates based on estimated revenue potential, competitive positioning, guest rating trajectories, and operational fit with Hotiday's model. This transforms property evaluation from a manual, intuition-driven process into a repeatable, data-informed methodology. **DECISION-SUPPORT PROTOTYPE:** An interactive dashboard that combines both models into a usable tool for Hotiday's acquisitions team, enabling faster and more accurate expansion decisions. These deliverables directly address Hotiday's core operational need and are designed for real-world deployment, not just academic exploration.

Innovation and Multidisciplinarity

INNOVATION: While revenue management tools are common in hospitality, predictive intelligence for asset acquisition is largely unexplored. Existing solutions optimize operations within a property; this project innovates by moving upstream — using ML to decide which properties to operate in the first place. Combining proprietary data from 120+ assets with public tourism indicators creates a unique dataset unavailable to purely academic approaches. **DISCIPLINES INVOLVED:** The project integrates four perspectives: (1) Computer Science / Data Engineering — data pipelines, feature engineering, ML models; (2) Management Engineering — business model analysis, KPI definition, economic assessment; (3) Urban Planning / Territorial Analysis — destination dynamics, infrastructure impact, tourism geography; (4) Design — intuitive decision-support interface. These perspectives are not additive but multiplicative: ML models are only valuable if grounded in economic logic, informed by territorial understanding, and delivered through usable tools.

External Institutions and Tutors

Institutions: NA

Principal tutors: NA

Smart Lighting in Museums: Development of an Innovative Light Sensing and Adaptive Control System for Palazzo Ducale in Mantua

Contact Information

Project type: Industry & Innovation oriented

Main contact: Raffaella Trocchianesi - Dept. Design Politecnico di Milano - Via Durando 10, 20158 Milan.

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Overall Project Objectives

The project aims to develop an integrated smart lighting system capable of synchronizing natural and artificial light within museum environments. In the Camera degli Sposi, natural light is managed through PDLC films applied to the historic windows, while artificial light is controlled through a Tuya-based system governing 40 profile spotlights dedicated to the frescoes. The core objective is the creation of a supervisory algorithm that enables PDLC films and spotlights to interact intelligently, adapting illumination to external conditions and ensuring stable, conservation-oriented lighting. Scientific methodological objectives include establishing a multidisciplinary model involving exhibit designers, lighting designers, and computer engineers, and defining a replicable methodology for museum applications. Technical objectives include designing, prototyping, and testing an adaptive lighting system that dynamically responds to daylight variations. Beneficiaries include cultural institutions—especially small and medium-sized museums lacking internal technical resources—and organizations that manage exhibition spaces.

General Project Description

This project proposes an innovative approach to museum lighting through the digitalization and engineering of an intelligent system capable of synchronizing natural and artificial light. The system integrates PDLC films, which modulate daylight transmission based on external conditions, and digitally controlled profile spotlights whose luminous flux can be adjusted in real time. A central supervisory algorithm will coordinate both components, enabling the luminous environment to behave as a responsive, adaptive organism. The project is situated within the broader context of museum exhibition design, where lighting represents a complex challenge—particularly in historic environments. Although the methodology is widely applicable, the system will be tested in a prestigious and highly demanding context: the Camera degli Sposi in Palazzo Ducale, Mantua. The motivation is to provide museums and designers with a flexible, intelligent tool capable of “dosing” light according to environmental conditions and conservation needs. Stakeholders include:

- Palazzo Ducale di Mantova — project context and testing environment
- Lighting manufacturers — providers of fixtures, PDLC films, and technical support
- Technology partners — support for sensor integration and control systems

Timeline and phases:

- M1–M2: Analysis of museum context, spatial conditions, artworks, exhibition apparatus, and lighting requirements
- M3–M5: Lighting system design and development of the supervisory algorithm
- M6–M8: Prototyping of the integrated lighting system
- M9–M10: In situ testing of the prototype in the Camera degli Sposi
- M11–M12: Refinements and additional testing
- M12–M13: Final report and dissemination of results

Expected Results

- Optimal lighting levels for fresco preservation, with reduced glare and energy consumption
- Reduced photodegradation and thermal load on artworks through dynamic control of natural/artificial light
- Continuous monitoring of the artwork’s condition through a historical database of the illuminance conditions to which it has been subjected
- Visual comfort and dynamic readability of the artwork, e.g., highlighting specific details for thematic tours
- Automated light management and monitored operation
- New commercial applications for partner companies through validation in a prestigious context

Deliverables:

- A mathematical model describing the interaction between natural and artificial light
- An adaptive control algorithm regulating PDLC films and spotlights in real time
- A

functional prototype installed and tested in the Camera degli Sposi •
 A replicable methodology (protocol) for smart lighting in museums
 • Documentation and guidelines for future implementations These results address the need for conservation-oriented, flexible lighting systems in historic museums, offering institutions a scalable and technologically advanced solution.

Innovation and Multidisciplinarity

The project is innovative because it integrates natural and artificial light through a unified intelligent system—something not currently available in museum lighting practice. The combination of PDLC films, sensor networks, and adaptive spotlight control introduces a new paradigm for heritage illumination. Disciplines involved include exhibit design, lighting design, computer engineering, optics, and conservation science. Their integration is essential: exhibit designers shape visitor experience, lighting designers define luminous quality, and engineers develop the technological infrastructure. Together, they create a systemic approach where perceptual, technical, and conservation needs converge.

External Institutions and Tutors

Institutions: • Palazzo Ducale in Mantua • Rimani (A Turin-based company specialized in the design and implementation of innovative lighting solutions, with a strong focus on art and cultural heritage.)
 • Gigi Events srl (Innovative Italian startup in the field of art and museum lighting technology)

Principal tutors: • Arch. Francesco Murano • francescomurano@yahoo.com
 • GIGI EVENTS S.R.L.

19

DATA CENTERS AS INCUBATORS OF INNOVATION ON SLOW, RESILIENT TERRITORIES

Contact Information

Project type: Industry & Innovation oriented

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Overall Project Objectives

The project aims to study how the proposal, design, and construction of data centers is increasingly impacting territories with slow decision-making processes and territorial structures, mostly in defense and with consolidated planning structures that are partly obsolete in their regulation. The project aims to highlight the contrast between the speed and innovative power of data centers and the firmness of the territories to defend consolidated values, in protection and in protection of already defined structures. The experience aims to find design, planning, and decision-making solutions capable of reflecting the immateriality of a data center's innovation, between artificial intelligence and data exchange, on the materiality of the contexts in which they are located. It therefore works on the image of buildings, their impact on environmental and settlement systems, and their ability to exchange innovation and information with the contexts in which they are located. The student will acquire the expertise of planning, drawing and building a data center, as a modern data exchange machine, in a box whose logic is not and is not yet particularly innovative

General Project Description

The project proposes to work towards the construction of a territorial strategy that welcomes the construction of a data center in a more conscious context and greater exchange between the established function and the already consolidated settlement system. Work on understanding how local decision-making processes are structured, between rules and oppositional conflicts, and what benefits building a data center brings. A classic case of disconnection between a place and what is built there emerges, which does not necessarily exchange benefits or resources with the place itself. Students will understand how to work on this contrast, but at the same time they will need to be able to open the data center box to understand what, within it, can be exchanged with the local context. We will start with concrete case studies, starting with the many data centers under design and construction around Milan. We will visit one of them and engage in dialogue with a private entity that develops and builds them (Engineering 2K) and the main private entity that promotes them (Amazon), both of whom are invited to participate in the project module. The project will work on this experience for a few months, depending on the school schedule. At the same time, a major design studio will be involved, very active in the field of planning processes for approving a data center, and specialized in their design. The student will thus be able to understand its genesis and read its outcome from multiple points of view

Expected Results

The student will learn the expertise to deal with complex and contrasting situations, and will learn to enter into complex decision-making processes, engaging with organizations, simulating tables of actors, and understanding how to transfer complex and specialized content (such as that expressed by a data center) to slower contexts and formats with different values. A toolkit of tools capable of reading, interpreting, and engaging in dialogue will be developed, but above all, we will understand how to use innovative and complex content to renew dialogue with the public and those responsible for planning decision-making processes. The close dialogue with private entities that are the main actors in the construction of data centers will be of great stimulus, importance, knowledge and visibility for the student

Innovation and Multidisciplinarity

Innovation is at the heart of the project. Talking about data centers means talking about the most recent urban planning function that adds to the other canonical and more consolidated ones in the area. Similarly, the relationship between a local context and the construction of a box that has exchange relationships with a non-local context is innovative, in which physicality becomes relative and connected to the speed of exchange processes. It will thus be possible to understand how to renew the language of dialogue with public bodies, with authorization dynamics, with consensus building, with relationships with the environment, soil, air

External Institutions and Tutors

Institutions: ENGINEERING 2K + AMAZON + THE BLOSSOM
AVENUE PARTNERS + COMUNE DI PREGNANA MILANESE (MI)

Principal tutors: Marco Dellavalle (TBAP)

How Can We Shape Waves? Learning from Nature to Control Sound in Water

Contact Information

Project type: Research oriented

Main contact: Prof. Marco Miniaci, Politecnico di Torino, Dipartimento di Ingegneria Strutturale, Edile e Geotecnica (DISEG), Corso Duca degli Abruzzi, 24, 10129, Torino (TO), Italia

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Overall Project Objectives

Wave propagation is strategically important across many applications, from medical ultrasound and acoustic communication to noise mitigation in marine ecosystems. In these contexts, the ability to manipulate sound is essential to reduce reliance on complex active or electronically controlled systems. Natural systems achieve remarkable wave manipulation through geometry, hierarchy and spatial organization, enabling filtering, redirection and attenuation via impedance grading, multi-scale scattering and local resonances. These functionalities emerge from morphology rather than technological complexity, providing a powerful structural paradigm for wave control. This project translates these bio-inspired principles into acoustic metamaterials – engineered architectures capable of tailoring wave propagation through geometry. Beyond conventional periodicity, quasi-periodic, graded and disordered patterns will be explored to achieve broadband and low-frequency performance in water. The project will design, fabricate and experimentally validate underwater metamaterial panels, uncovering the governing physical mechanisms and translating them into scalable engineering solutions.

General Project Description

The project brings together complementary expertise in wave physics, structural mechanics, computational modelling (including AI-based techniques), materials engineering and experimental acoustics. The industrial partner Phononic Vibes will contribute with its expertise in manufacturability, scalability and potential technological transfer. The project will be structured into the following phases (RE = Required expertise, COO = Coordination):

M1–2 – State of the Art. Review of underwater acoustics across sonic/ultrasonic regimes and of acoustic metamaterials. This phase includes seminars on wave physics, bio-inspired architecture and multi-physics modelling (e.g., COMSOL) to establish a shared knowledge base. Target functionalities (mode conversion, isolation, redirection, focusing) and measurable performance indicators will be defined. RE: physics, mechanical/aerospace engineering, applied mathematics, materials science, computational modelling. COO: All partners.

M2–4 – Preliminary Modelling. Identification of relevant natural patterns and their translation into candidate metamaterial architecture. Analytical and numerical studies will investigate dispersion, impedance matching and fluid–structure mode coupling in water, providing initial design guidelines. RE: physics, mechanical/aerospace engineering, applied mathematics, materials science, computational modelling. COO: POLITO.

M4–6 – Numerical Optimization and Design Refinement. Parametric studies and multi-objective optimization of geometric and material parameters to enhance isolation, redirection and focusing performance. RE: computational modelling, optimization, structural mechanics. COO: POLIMI.

M7–9 – Prototype Fabrication. Manufacture of selected designs using additive manufacturing and precision machining, with input from the industrial partner on feasibility, tolerances and scalability. Activities will be carried out at DMEC–POLIMI and Phononic Vibes facilities. RE: materials engineering, manufacturing technologies, industrial expertise. COO: POLIMI, Phononic Vibes.

M9–12 – Experimental Characterization and Iteration. Experimental validation through laser Doppler vibrometry and underwater acoustic testing in dedicated tanks covering both low (<20 kHz) and high (>200 kHz) frequency regimes. Results will be compared with simulations, enabling iterative design–fabrication–testing cycles for performance refinement. RE: acoustics, vibration mechanics, experimental methods. COO: POLITO, POLIMI.

Expected Results

The expected outcome is the development and validation of scalable metamaterial-based devices for underwater sound control, targeting applications from medical ultrasound technologies to noise mitigation in marine ecosystems. The project will deliver: 1) a validated design framework for geometry-driven manipulation of acoustic waves in water, including modelling tools; 2) one or more demonstrator metamaterial panels, designed, fabricated and experimentally characterized in dedicated underwater facilities, operating within selected frequency bands; 3) analysis of representative application cases, such as ultrasonic focusing in biological tissue for sensing or therapy, underwater noise mitigation from offshore infrastructures. These outcomes address the growing demand for effective and scalable underwater sound-control strategies, providing the industrial partner with validated, manufacturable design principles. In parallel, the project will foster interdisciplinary expertise at the interface of wave physics, structural mechanics, numerical modelling, and experimental acoustics, training students to bridge fundamental research and real-world applications.

Innovation and Multidisciplinarity

Innovation. While metamaterials are extensively studied in air and solids, their systematic development in underwater environment remains limited. The innovation of this project lies in translating bio-inspired architectures into scalable underwater metamaterial panels and in identifying the governing physical mechanisms – dispersion tailoring, impedance matching, and fluid–structure mode coupling – that enable efficient wave control over broad frequency ranges. This approach moves beyond isolated demonstrations toward a unified, geometry-driven design framework for underwater acoustics. **Multidisciplinarity.** The project integrates wave physics, structural mechanics, applied mathematics, multiphysics numerical modelling, and experimental acoustics. Physics establishes the theoretical description of wave propagation; mechanics and materials science enable structural implementation; computational engineering supports optimization and performance mapping; and laboratory testing in underwater facilities provides validation. Their coordinated integration is essential to bridge fundamental understanding with manufacturable devices and application-oriented performance.

External Institutions and Tutors

Institutions: Phononic Vibes Srl - Via Europa, 28, 20083 Vigano
Milano

Principal tutors: Luca Sangiuliano (luca.sangiuliano@phononicvibes.com
- Via Europa, 28, 20083 Vigano Milano)

Additional tutors: Marco Antonacci (marco.antonacci@phononicvibes.com
- Via Europa, 28, 20083 Vigano Milano), Luca D'Alessandro (luca.dalessandro@phononicvibes.com
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Production Process Optimisation with AI

Contact Information

Project type: Industry & Innovation oriented

Main contact: Ferdinando Loiacono - AI Project Lead - Sedamyl - Via Monviso 18, 12037 Saluzzo – CN

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Overall Project Objectives

SEDAMYL is a leading manufacturer of starches, sweeteners, proteins, and alcohol, operating production plants in Saluzzo (Italy) and Selby (UK). As part of its digital transformation strategy, SEDAMYL is focusing on the application of Artificial Intelligence to deliver tangible operational benefits. This proposal addresses a strategically significant initiative: the selection of a software partner to develop customised AI solutions capable of monitoring and dynamically adjusting the operating parameters of starch and gluten dryers in real time, with the aim of improving process reliability and production yields. The project explores advanced levels of automation and digital innovation that are new to the sector and have the potential to generate meaningful cost and sustainability efficiencies. The involvement of ASP students broadens the scope beyond software development to include the effective integration of digital and AI tools within existing industrial operations, ensuring alignment between technology, processes, and organisational capabilities.

General Project Description

SEDAMYL will collaborate with a specialized software partner to design and implement machine learning algorithms capable of automatically controlling the starch and gluten dryers across both production sites. The primary technical objective is to enhance starch and gluten moisture control, with the aim of increasing final product

moisture levels by approximately 1–2 percentage points without interrupting production. Current operating practices intentionally run below the maximum moisture threshold to minimize the risk of unplanned stoppages. By enabling more precise and responsive control, the project seeks to improve operational yields and unlock associated cost and sustainability efficiencies. The planned activities align with the Data to Action paradigm, whereby operational data are algorithmically transformed into the capability to actively intervene in and optimize physical industrial processes. This initiative represents a fully applied innovation project, characterized by a recursive, collaborative, and agile development approach, specifically focused on converting data insights into actionable operational capabilities. ASP students will play an active role across multiple dimensions of the project. Their contributions will include support in supplier selection, technical and business validation of the underlying assumptions, and assistance with change management related to the introduction of advanced digital tools within existing industrial processes. In addition, students will participate in the development of the software application and may also explore and propose alternative pilot use cases. The project timeline for the development of the software application is the second half of 2026, followed by its implementation within operational processes in the first half of 2027, with full organizational adoption. The full operational, cost, and sustainability benefits are expected to be measurable during 2027.

Expected Results

This project is innovative in the industry since generally all efficiency improvements are focused on process improvements while the objective is to integrate innovative digital technology to aid plant production and improve operational yield. The main tangible result will be the improved yield compared to historic values but the other important result from SEDAMYL is to learn to integrate innovative digital technology into production processes of commodity products, grinding multiple hundred thousand tons of wheat per year. Our objective to involve ASP in our project is to open our doors to innovation and inspire and be inspired by young talented students.

Innovation and Multidisciplinarity

The project adopts a collaborative, integrated, and iterative development approach, bringing together cross-sector and cross-disciplinary expertise, including data engineering, data science, business analysis, finance, and domain specialists. A systematic cross-validation process

ensures that hypotheses and quantitative results are consistently validated from physical, engineering, and business perspectives. The Business Scope phase defines objectives, constraints, and context, guiding project setup and development. Data Engineering focuses on acquiring, preparing, and understanding the data required to train AI models. Data Insight assesses data quality and relevance against model objectives. Data Science involves the iterative design, testing, and refinement of customised, high-performance AI models. Finally, Business Insight ensures that developed solutions meet functional, quality, and ethical requirements through structured testing in controlled environments, supporting system reliability and user acceptance prior to deployment.

External Institutions and Tutors

Institutions: External software development company not yet appointed

Principal tutors: Ferdinando Loiacono ferdinando.loiacono@sedamyl.com

Additional tutors: Luigi Alesso luigi.alesso@sedamyl.com

Powering IoT with Indoor Light: Integrated Energy Harvesting and Storage Device for Autonomous Sensors

Project type: Research oriented

Contact Information

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Overall Project Objectives

The project aims to develop a compact device capable of both harvesting and storing energy from indoor light sources to power low-consumption Internet of Things (IoT) devices such as distributed sensors. The system integrates a dye-sensitized solar cell (DSSC) and a supercapacitor on a single substrate, enabling simultaneous energy generation and storage without external power management circuits. This integrated approach addresses a major limitation in IoT deployment: the need for battery replacement or maintenance in large networks of sensors. By enabling autonomous power supply from ambient light, the proposed solution can significantly reduce maintenance costs and environmental impact. A multidisciplinary approach combining materials science, electrochemistry, electrical engineering, and system design is essential to optimize the device architecture and validate it in real-world scenarios. Potential beneficiaries include IoT technology companies, sensor manufacturers, energy storage developers, and sectors relying on distributed sensing

systems such as smart buildings, healthcare monitoring, agriculture, and energy management.

General Project Description

The rapid expansion of the Internet of Things is creating an increasing demand for reliable and maintenance-free energy sources for distributed sensors. In many IoT applications—such as smart buildings, healthcare monitoring, wearable electronics, and environmental sensing—devices operate indoors under low light conditions, where conventional photovoltaic technologies are less efficient and batteries require frequent replacement. The project proposes the development of a hybrid two-electrode device capable of both harvesting and storing energy from indoor light. The system integrates a dye-sensitized solar cell and a supercapacitor fabricated through a unified process on the same substrate. This monolithic architecture eliminates the need for external power management circuits and enables a compact, scalable, and ready-to-use energy module. A functional laboratory demonstrator already exists (TRL 4–5), showing promising performance under indoor lighting conditions. The project aims to advance the technology to TRL 6 through optimization, integration, and validation in a specific IoT use case. The project will involve collaboration between research institutions and industrial stakeholders interested in energy harvesting and IoT technologies. Research activities will be carried out primarily at the Italian Institute of Technology (IIT) and Politecnico di Torino. Potential industrial partners include companies active in photovoltaic materials, energy storage technologies, and IoT systems. The work plan is structured into four main phases: Definition of the target application and design of the optimized prototype. Optimization of materials and fabrication processes for the DSSC and the supercapacitor. Integration of the components into a single device and laboratory testing. Validation of the system under realistic operating conditions. The project is expected to last approximately 8–10 months and will result in a fully integrated prototype capable of generating, storing, and retaining energy from indoor light sources.

Expected Results

The main expected outcome of the project is a fully integrated energy harvesting and storage device optimized for indoor IoT applications. Key deliverables include: An optimized prototype integrating a dye-sensitized solar cell and a supercapacitor on a single substrate. Experimental validation of the device under indoor lighting con-

ditions. Performance characterization in terms of energy density, power output, and self-discharge behavior. Demonstration of the device powering a representative low-power IoT sensor. A technical assessment of scalability and manufacturability. These results will address a critical challenge faced by IoT companies: ensuring reliable and maintenance-free power supply for distributed sensors. The proposed device can significantly reduce reliance on batteries and enable long-term autonomous operation of sensing networks. The project may also support future technology transfer activities, including patent development, industrial partnerships, or licensing opportunities with companies operating in the fields of energy harvesting, photovoltaics, and energy storage.

Innovation and Multidisciplinarity

The project is innovative because it proposes a monolithic device that simultaneously harvests and stores energy from indoor light, eliminating the need for separate photovoltaic modules, storage units, and external power management electronics. Current commercial solutions typically combine solar cells with external capacitors or batteries, resulting in larger, more complex systems. The proposed architecture integrates a dye-sensitized solar cell with a supercapacitor in a two-electrode configuration, enabling direct energy conversion and storage on a single substrate. In addition, the system incorporates mechanisms designed to minimize self-discharge, allowing energy to be retained even when light is absent. The project requires the integration of several disciplines: Materials science, for the development and optimization of functional materials for DSSCs and supercapacitors. Electrochemistry, for understanding and controlling charge transfer processes and self-discharge mechanisms. Electrical engineering, for device architecture design and performance evaluation. Energy systems engineering, for integration into real IoT applications.

External Institutions and Tutors

Institutions: istituto italiano di tecnologia

Principal tutors: Isabella Poli, IIT, isabella.poli@iit.it

SWARM-SHELL Swarm Robotics for Additive Manufacturing of Shell Structures

Contact Information

Project type: Industry & Innovation oriented

Main contact: Valentina Sumini, Politecnico di Torino, Department of Structural, Geotechnical and Building Engineering (DISEG), Corso Duca degli Abruzzi, 24, Torino

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Overall Project Objectives

This project investigates the potential of swarm robotic systems for the distributed additive manufacturing of shell structures. The research explores how multiple autonomous robots can cooperate to deposit material and fabricate structurally efficient geometries derived from computational design methods and bio-inspired principles. By coordinating decentralized robotic agents, the system aims to enable scalable construction processes capable of adapting to complex spatial and environmental conditions. Shell structures are particularly suitable for this approach due to their structural efficiency and ability to span large areas with minimal material use. The project will examine algorithms for swarm coordination, robotic path planning, and real-time construction feedback, integrating digital design tools with robotic fabrication strategies. Beyond terrestrial applications in automated construction and advanced manufacturing, the research also considers implications for space exploration. In extraterrestrial environments, robotic swarms could autonomously fabricate infrastructure before human arrival, supporting the development of resilient habitats in extreme and remote contexts.

General Project Description

Context and Motivation Advances in computational design, robotic fabrication, and artificial intelligence are transforming how complex structures are conceived and built. Swarm robotics introduces a decentralized paradigm in which multiple autonomous agents cooperate to fabricate structures through distributed additive manufacturing. This approach enables scalable and adaptive construction strategies that overcome many limitations of centralized building systems. Shell structures are an ideal case study due to their structural efficiency, lightweight properties, and ability to span large areas with minimal material use. These characteristics make them particularly relevant for extreme environments such as the Moon, where construction must rely on autonomous systems and in-situ resource utilization (ISRU). The project investigates how coordinated robotic swarms can fabricate structurally optimized shell habitats, with potential applications in both advanced terrestrial construction and future space infrastructure.

Stakeholders and Roles The project involves a multidisciplinary collaboration between academia, research laboratories, and industry. Researchers from Politecnico di Torino and Politecnico di Milano will develop AI-driven design methodologies for shell structures, focusing on computational design, swarm coordination, and additive manufacturing constraints. Their work will explore how decentralized robotic systems translate digital geometries into cooperative fabrication strategies and enable new forms of automated construction, including space habitats. The MIT Responsive Environments Group will contribute expertise in responsive systems and interactive robotic environments, supporting decentralized coordination strategies and adaptive fabrication logics. The industrial partner Habitatto will provide insights into autonomous construction for lunar habitats, including ISRU, AI-driven operations, and additive manufacturing technologies.

Timeline and Key Phases The project is structured in three phases: background research and development of computational models for shell geometries and swarm coordination; simulation and testing of swarm robotic behaviors, including path planning and cooperative fabrication workflows; and integration of computational design with robotic control into a digital prototype workflow to evaluate the feasibility and architectural implications of swarm-based additive manufacturing.

Expected Results

The project will produce conceptual and computational outcomes demonstrating the feasibility of swarm robotics for the additive

manufacturing of shell structures. First, the research will develop a computational design framework for structurally optimized shell geometries suitable for robotic fabrication, integrating AI-driven design methods with geometric and structural criteria tailored to distributed additive manufacturing. Second, the project will create a simulation environment for swarm robotic construction, exploring coordination algorithms, path planning, and cooperative material deposition workflows. Simulations will evaluate how multiple robots can collaboratively fabricate complex geometries while maintaining structural continuity and efficiency. Another outcome will be the definition of design guidelines for robotic additive manufacturing of shell structures, addressing geometry, structural performance, scalability, and fabrication logic. Key deliverables will include a computational design workflow, swarm robotics simulation models, and a conceptual architectural scenario illustrating the potential of this technology for autonomous construction systems.

Innovation and Multidisciplinarity

The project proposes an innovative approach to architectural construction by integrating swarm robotics, AI-driven computational design, and additive manufacturing. Unlike conventional systems based on centralized machines or single robotic arms, it explores a distributed fabrication model in which multiple small robots cooperate to build structurally efficient shell structures. By linking computational design with swarm robotic logic, the research investigates how optimized geometries can be translated into cooperative fabrication strategies, enabling scalable and adaptive construction. This approach is particularly relevant for extreme environments on Earth and in space, where autonomous systems and in-situ resource utilization are essential. The project is inherently multidisciplinary, combining architecture and computational design with robotics, artificial intelligence, and digital fabrication. This integration enables the development of decentralized robotic construction strategies that connect design, robotic systems, and fabrication processes within a unified research framework.

External Institutions and Tutors

Institutions: MIT Media Lab; HABITATO

Principal tutors: Prof. Joseph Paradiso, joep@media.mit.edu, MIT Media Lab, Responsive Environments; Dr. Johannes Lierfeld, johan-neslierfeld@gmail.com, HABITATO / LUNAR HABITATO

Il Motore Digitale dell'Organizzazione del Futuro

Contact Information

Project type: Business oriented (entrepreneurship & high tech start-ups)

Main contact: Marco Stroppiana, Responsabile Servizio Organizzazione - Banca di Asti

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Overall Project Objectives

Il progetto introduce un'architettura innovativa di trasformazione tecnologica che supera le iniziative esistenti unificando livelli eterogenei di maturità dei processi in un modello di adozione digitale scalabile e controllato. L'innovazione deriva dall'integrazione di reference architecture, standard di interoperabilità e framework di misurazione delle performance, che consentono un deployment prevedibile, misurabile e sicuro di tecnologie avanzate come motori di decisione creditizia basati su AI, moduli di robotic process automation e agenti conversazionali arricchiti da knowledge graph. Obiettivi di Progetto: -Realizzazione di un service desk intelligente, supportato da AI; -Modernizzazione delle soluzioni di Office automation, con migrazione verso: 1) una data platform aziendale dotata di controlli automatici di qualità, tracciamento della lineage e processi di ingestion standardizzati 2) utilizzo di strumentazione low code/no code

General Project Description

Il progetto ha l'obiettivo di definire un'architettura sistemica di trasformazione tecnologica per Banca di Asti, fondata su una strategia di adozione strutturata e su una roadmap esecutiva di tipo ingegneristico. L'iniziativa mira a costruire un modello operativo scalabile, capace di armonizzare differenti livelli di maturità dei processi e di

supportare un deployment controllato delle tecnologie digitali avanzate. Il framework proposto integra standard di interoperabilità e modelli di misurazione delle performance per ottimizzare il throughput operativo, ridurre la variabilità dei processi e minimizzare la latenza end-to-end dei servizi. L'elemento innovativo risiede nella formalizzazione di un blueprint multidimensionale che coordina in modo coerente i livelli infrastrutturali, applicativi, organizzativi e di data governance. La realizzazione del progetto richiede competenze multidisciplinari che includono systems engineering, architetture distribuite, data platform engineering, pipeline di machine learning, process mining e metodologie di organizational design. Ciò consente l'implementazione di motori di decisione creditizia basati su AI, soluzioni di robotic process automation e agenti conversazionali supportati da knowledge graph, nel rispetto dei requisiti di affidabilità, scalabilità, sicurezza e dei framework di governance ITIL/COBIT. Il progetto prevede inoltre la progettazione e la validazione di casi pilota operativi finalizzati a dimostrarne fattibilità e impatto, tra cui: – un service desk basato su AI, integrato con pipeline di NLU, knowledge graph e classificazione automatica degli incident; – workflow automatizzati sviluppati con strumenti no-code/low-code, supportati da repository documentali centralizzati, feature store e modelli di apprendimento supervisionato; – la re-ingegnerizzazione delle soluzioni di Office automation tramite migrazione verso una data platform aziendale con controlli automatici di qualità del dato, tracciamento della lineage e framework di ingestion standardizzati. Ogni pilota produrrà KPI quantitativi (riduzione dei tempi di ciclo, tasso di errore, metriche di affidabilità), abilitando la valutazione del ROI e della scalabilità. I beneficiari principali saranno le unità operative, l'organizzazione nel suo complesso e i clienti finali, che sperimenteranno servizi più continui e a minore latenza.

Expected Results

Risparmio di circa 2.000.000 € sui costi di running su base annua, accompagnato dalla riallocazione delle risorse.

Innovation and Multidisciplinarity

Il progetto è innovativo perché non introduce singole tecnologie, ma un'architettura integrata che governa in modo unitario adozione, controllo e scalabilità dell'innovazione digitale. Integra architetture di riferimento, standard di interoperabilità e KPI per rendere prevedibile e misurabile l'uso di AI, RPA e soluzioni low code/no code. Coinvolge discipline complementari – architetture IT, data gover-

nance, AI engineering, process mining e organizational design – la cui integrazione è necessaria per allineare tecnologia, processi e organizzazione, trasformando l’innovazione in un modello operativo sostenibile e scalabile.

External Institutions and Tutors

Institutions: Nessuno

Principal tutors: Nessuno

25

Semi professional high-speed-cooking oven

Contact Information

Project type: Business oriented (entrepreneurship & high tech start-ups)

Main contact: Luana Spagnoli - Electrolux Professional - Viale Treviso, Pordenone

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Overall Project Objectives

Develop a compact semi-professional high-speed oven, maximum 3 trays, designed for rapid reheating of pre-cooked foods, panini warming, and simple bakery finishing (e.g., small pastries, focaccine, basic laminated products). The oven must be ultra-compact, suitable for table-top installation, and ideal for environments such as service stations, cafés, convenience stores, and foodservice kiosks where speed, quality, and minimal operator skill are essential. Deliver an optimal combination of speed, quality, and ease of use, enabled by an integrated set of thermal technologies: Microwave heating to fast heating to the core; Intelligent impingement / air-jet system for rapid browning, crisping, and uniformity; Steam-Assist to prevents dryness and improves bakery regeneration

General Project Description

This project aims to develop the analytical and experimental proof of concept (TRL 3) of a compact high-speed thermal system designed for modern foodservice environments such as service stations, cafés, and convenience stores. These contexts require fast and repeatable heating of panini, pre-cooked items, and simple bakery products, but typically operate in limited spaces and with non-specialized staff, making traditional thermal equipment unsuitable. To address this need, the project explores a multi-technology approach combining microwave volumetric heating, intelligent impingement airflow, and

steam-assist humidity control within a small table-top cavity. This concept aims to demonstrate that high-performance, scalable heating systems can be achieved even in highly constrained environments.

Stakeholders and Roles The project involves an interdisciplinary team of ASP students: Thermal, Mechanical and Aerospace Engineering students will model heat transfer, airflow distribution, and microwave propagation, defining the core physical feasibility; Electronics and Automation students will design control logic, sensing strategies, and communication protocols for the modular architecture. Computer Science and Interaction Design students will develop UI prototypes and software logic for multi-cavity coordination. Industrial Design students will address ergonomic, spatial, and integration constraints. Food Technology students will validate heating curves, moisture behavior, and product quality. Academic supervisors will guide methodological rigor, while potential industrial partners may provide benchmarks and application scenarios.

Timeline and Key Phases Phase 1 - Context & Requirements: definition of functional needs, KPIs, and constraints. Phase 2 - Modelling & Concept Development: CFD airflow simulations, microwave field modelling, steam-assist studies, and architecture definition. Phase 3 - Experimental Validation: bench-scale tests of individual subsystems and preliminary integration. Phase 4 - Proof-of-Concept Demonstration: validation of thermal behavior, and feasibility for TRL 3. Phase 5 - Final Consolidation: report, presentation, and recommendations for advancement to TRL 4.

Expected Results

The project will deliver a TRL-3 proof of concept, demonstrating the feasibility of combining microwave heating, intelligent impingement airflow and steam-assist in a compact table-top system. Expected results include validated thermal and airflow models, microwave energy-distribution simulations and humidity-control strategies showing how the three technologies operate synergistically to ensure fast, uniform and moisture-stable heating. A second key outcome is the definition of a modular Master/Slave architecture, with specifications for communication protocols, coordinated control logic and an interface prototype capable of managing multiple cavities through a single UI. The project will also produce bench-scale experimental results confirming heating curves, browning uniformity and product quality on real food items. Additional deliverables include a preliminary mechanical concept for the compact oven and a UX/UI mock-up enabling user-friendly operation in constrained environments.

Innovation and Multidisciplinarity

This project explores the integration of microwave heating, intelligent impingement airflow and steam-assist to achieve fast, uniform and moisture-controlled heating in a compact cavity. It also introduces a modular Master/Slave architecture that increases capacity without duplicating interfaces or control systems, reducing cost and operational complexity. The work involves multiple disciplines: thermal and fluid dynamics for heat and airflow modelling; microwave/RF engineering for energy distribution; mechanical and industrial design for a safe and compact structure; electronics and control systems for sensing, actuation and module communication; software and UX design for a unified interface; and food technology to validate product performance. These disciplines are all necessary, as the challenges of speed, uniformity, moisture stability, space constraints and usability cannot be solved by a single field. Their integration creates a coherent interdisciplinary approach that enables an innovation impossible through isolated expertise.

External Institutions and Tutors

Institutions: Electrolux Professional

Principal tutors: Luana Spagnoli, luana.spagnoli@electroluxprofessional.com,
Electrolux Professional

Additional tutors: Arianna Bozzato, arianna.bozzato@electroluxprofessional.com,
Electrolux Professional

RANGE EXTENDER FOR SMALL UAV SYSTEMS

Contact Information

Project type: Industry & Innovation oriented

Main contact: Riccardo Zich
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Overall Project Objectives

The project aims to design and validate the concept of a lightweight range extender for small unmanned aerial vehicles below 25 kg, based on a miniaturized internal combustion engine coupled with an electric power generation system. The main objective is to increase flight endurance while preserving compliance with strict constraints on mass, volume, safety, vibration, thermal management, and integration with the airframe. The innovation lies in combining propulsion, energy conversion, mechanical integration, and control into a compact hybrid solution suitable for next-generation UAV missions. A multidisciplinary approach is essential, involving aerospace, mechanical, energy, electrical, and control engineering, as well as system integration and industrial design. The expected beneficiaries are UAV manufacturers, operators of long-endurance aerial platforms, and end users in surveillance, inspection, environmental monitoring, civil protection, and security applications. The project may also generate broader benefits for lightweight hybrid power systems in advanced air mobility and unmanned systems.

General Project Description

Small unmanned aerial vehicles below 25 kg are increasingly used in inspection, monitoring, civil protection, security, and environmental applications. Their main limitation is endurance: battery-based platforms are often constrained by short flight times, which reduce mission range, payload flexibility, and operational continuity. This

project addresses that limitation by exploring the design of a compact range extender based on a miniaturized internal combustion engine coupled with an electric generation unit, to be integrated into a lightweight UAV platform. The project is motivated by the need for a practical and scalable solution able to extend mission duration without moving to much larger and more regulated aerial platforms. The challenge is not only technological, but systemic: the range extender must be designed as an integrated subsystem, balancing power output, weight, vibrations, thermal loads, safety, acoustic impact, maintainability, and compatibility with UAV architecture and mission requirements. The main stakeholders are: academic tutors from Politecnico di Milano and Politecnico di Torino, who will guide the methodological and scientific aspects; external industrial partners and technology providers, who will contribute application needs, system constraints, and engineering feedback; and the ASP student team, which will carry out the multidisciplinary development work. Potential end users include UAV manufacturers, service operators, and organizations active in surveillance, infrastructure inspection, environmental monitoring, and emergency management. The work is expected to develop in four main phases. The first phase focuses on state-of-the-art analysis, benchmarking of existing hybrid and range-extender solutions, and definition of mission and system requirements. The second phase addresses concept generation and preliminary system architecture, including engine-generator selection, energy balance, mass budget, and integration constraints. The third phase develops modeling and design activities on mechanical integration, thermal management, vibration mitigation, control logic, and safety strategies. The final phase is devoted to performance assessment, trade-off analysis, and definition of a feasible development roadmap, including possible prototyping scenarios, validation criteria, and industrialization perspectives.

Expected Results

The project is expected to deliver a feasibility-oriented concept for a lightweight range extender for UAVs below 25 kg, supported by a clear set of technical outputs. The main deliverables will include: a state-of-the-art and benchmarking analysis of existing hybrid and range-extender solutions; a definition of mission and system requirements; a preliminary system architecture for the engine-generator unit and its integration into a small UAV platform; and a trade-off analysis covering power output, mass, thermal behavior, vibrations, safety, and maintainability. Additional results will include a mass and energy budget, preliminary integration and control logic, and a

development roadmap identifying the most promising options for future prototyping and testing. These outputs address a concrete need expressed by UAV operators and technology stakeholders: extending mission endurance without moving to larger, more complex, and more regulated aerial platforms. The project will therefore provide both a technical basis and a decision-support framework for organizations interested in long-endurance UAV solutions for inspection, monitoring, security, and environmental applications

Innovation and Multidisciplinarity

The project is innovative because it addresses a key limitation of small UAVs — short endurance — through a compact range extender concept specifically designed for platforms below 25 kg. Existing solutions are usually developed for larger aerial systems or as standalone propulsion components, while this project focuses on the integrated design of a miniaturized internal combustion engine, electric generation, power management, and UAV compatibility under strict mass, volume, safety, and reliability constraints. The project combines aerospace engineering, mechanical engineering, energy and power systems, electrical and control engineering, and system design. Their integration is essential because the range extender is not an isolated component, but a tightly coupled subsystem whose feasibility depends on the balance between propulsion needs, structural constraints, thermal behavior, vibrations, control logic, and operational requirements. The different disciplines complement each other by linking flight integration, mechanical layout, energy efficiency, power management, and practical feasibility into a single credible concept for future long-endurance UAV applications.

External Institutions and Tutors

Institutions: SUPAIR S.R.L. – aerospace technology company developing advanced aerial mobility solutions and lightweight UAV platforms; involved in the project as external industrial partner for system requirements, engineering constraints, and application-oriented feedback. Address: C.so Castelfidardo 30/A, 10129 Torino, Italy VAT: IT13156040019 Email: info@supair.it

Principal tutors: Angelo Lerro; angelo.lerro@supair.it ; SUPAIR S.R.L., CEO & Co-founder.

27

Scaling Hybrid storage for Innovative Future energy Technologies (SHIFT)

Contact Information

Project type: Research oriented

Main contact: Pietro Zaccagnini - Politecnico di Torino - Corso Duca degli Abruzzi 24, 10129, TO
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Overall Project Objectives

SHIFT objective is to explore how sustainable hybrid capacitor (HC) technologies can move from laboratory prototypes to scalable, sustainable and deployable energy storage solutions. While HCs offer exceptional power density and long cycle life, their transition toward large-scale applications remains limited by challenges related to materials sustainability, manufacturing scalability, and system integration. SHIFT aims to bridge this gap by combining multidisciplinary approaches. Specifically, SHIFT will: (O1) investigate sustainable materials and fabrication strategies for HC devices; (O2) experimentally characterize their electrochemical performance and reliability; (O3) analyze their behavior through modelling and ageing studies; (O4) explore their integration with other energy storage technologies and energy systems; (O5) assess the scalability and environmental implications of their production. By connecting materials research with system-level applications, SHIFT will provide students with a multidisciplinary perspective on energy storage innovation while generating insights that can support the development of next-generation sustainable energy storage technologies.

General Project Description

The global transition toward renewable energy and electrified mobility is accelerating the need for fast, efficient, and sustainable energy storage technologies. While batteries dominate long-term storage, hybrid capacitors are emerging as a key enabling technology for applications requiring high power density, rapid charging, and long cycle life, such as grid stabilization, renewable integration, electric mobility, and smart electronics. However, one of the major barriers preventing their widespread adoption is scalability: promising laboratory prototypes often struggle to transition into manufacturable, reliable, and sustainable devices. Bridging this gap between materials innovation and real-world deployment is a multidisciplinary challenge involving electrochemistry, materials science, electrical engineering, and energy systems modelling. SHIFT aims to explore how sustainable hybrid capacitors (HC) can be designed, fabricated, tested, and scaled up for real energy applications. The project will combine expertise from two leading research environments: (i) Politecnico di Torino (PoliTO) will contribute expertise in electrochemical characterization and device fabrication, focusing on innovative materials and sustainable manufacturing strategies. (ii) Politecnico di Milano (PoliMI) will contribute expertise in device testing, external hybridization, modelling, and ageing, enabling system-level analysis and long-term performance evaluation. Students will work in interdisciplinary teams to explore the full value chain of HC, from materials and device fabrication to system integration and sustainability assessment. Timeline and phases: P1 – Exploration & System Framing. Students analyze the state of the art in HC, identify promising sustainable materials and define target use cases. P2 – Device Design & Fabrication. Experimental work at PoliTO on electrode materials, electrochemical characterization, and fabrication of prototype devices. P3 – Testing, Modelling & Hybridization. Performance testing, ageing analysis, and system modelling at PoliMI, including integration scenarios with batteries or energy systems. P4 – Scaling & Sustainability Assessment. Students evaluate scalability, environmental impact, and potential industrial deployment pathways. The project aims to provide students with a hands-on experience in translating cutting-edge research into scalable energy technologies, preparing them to contribute to the future of sustainable energy systems.

Expected Results

SHIFT will deliver a comprehensive interdisciplinary assessment of scalable HC technologies and their role within future energy systems.

Expected output includes: (D1) Prototype HC devices based on sustainable materials and fabricated using scalable approaches. (D2) Electrochemical performance evaluation, including power density, energy density, efficiency, and cycle life. (D3) System-level modelling of HC integration with batteries or renewable energy systems. (D4) Ageing and reliability analysis, providing insights into long-term device performance. (D5) Scalability assessment, identifying key challenges and opportunities in moving from laboratory prototypes to industrial manufacturing. (D6) Preliminary sustainability evaluation, including environmental impact considerations and material choices. These results will help the involved groups explore new strategies for bridging the gap between materials research and deployable energy storage technologies. More broadly, the project will generate knowledge useful for future energy storage solutions supporting renewable integration, electrification, and resilient energy systems, addressing key challenges of the energy transition.

Innovation and Multidisciplinarity

SHIFT's innovativeness relies in addressing a critical but often overlooked challenge in energy storage research: the transition from promising laboratory materials to scalable and deployable devices. While many studies focus either on materials discovery or system integration, this project explicitly connects the entire innovation chain, from electrode materials to device performance and system-level applications. The project integrates multiple disciplines: materials science and chemistry to design sustainable electrode materials and characterize their behavior; electrical and electronic engineering to develop device architectures and testing protocols; energy engineering and modelling to evaluate how HCs interact with batteries and renewable systems; and sustainability and environmental analysis to assess the environmental implications of materials and production processes. The integration of these perspectives is essential because energy storage technologies must simultaneously meet performance, reliability, scalability, and sustainability requirements.

External Institutions and Tutors

Institutions: Italian Institute of Technology - Centre for Sustainable and Future Technologies - Turin

Principal tutors: Adriano Sacco - adriano.sacco@iit.it - Italian Institute of Technology

Rural Proximities. Designing Living Systems for Food, Energy and Multispecies Coexistence

Contact Information

Project type: Social oriented (social impact and engagement)

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Overall Project Objectives

In small and peripheral territories, infrastructures supporting everyday life, such as food production, energy generation, water management, housing and local economies are often fragmented and disconnected. These systems can be reimagined as integrated living infrastructures capable of supporting ecological regeneration, local development and new forms of community, functioning as nodes of broader territorial systems. This project investigates how ecological and socio-technical practices can be integrated into territorial systems connecting food, energy, water cycles, local economies, and multispecies coexistence. The agroecological farm Ca'Mariuccia in the Monferrato area will serve as a prototype and living lab. Through the analysis of its activities, including agroecology, renewable energy, water management, spatial and service models of sustainable living, the project adopts a multidisciplinary approach combining design, environmental studies, territorial planning, energy systems and social innovation. The research aims to support rural communities, local producers, and public institutions engaged in developing new models of territorial regeneration in small and remote places.

General Project Description

How can an agroecological farm evolve into a regenerative rural hub integrating productive, educational and living functions within sustainable economic and governance models? Small and remote territories are increasingly recognised as important contexts for innovation. While facing depopulation, economic fragility and limited access to services, these areas often experience forms of “relational remoteness”, where innovation and knowledge exchange remain weak despite the presence of strong ecological and cultural resources. At the same time, emerging ecological practices and new forms of distributed living and working are opening opportunities to rethink these territories in terms of rural proximities, understood as relational place-based ecosystems. Within this context, Ca’Mariuccia is an agroecological farm in the Monferrato area, managing about 50 ha of woods, water systems, pastures and cultivated land, combining agricultural production, animal breeding and food production with hospitality and educational activities while experimenting with ecological practices (e.g.: renewable energy, water management, landscape regeneration). Embedded in an active network of local producers, cooperatives, associations, municipalities and research organisations, Ca’Mariuccia provides a concrete environment to activate, test and monitor social and technological innovation practices, as a collaborative platform where multidisciplinary knowledge and resources converge.

1. Exploration and framing RQ1: What spatial, social and technological components define a regenerative rural hub? Tasks: Desk research and analysis of selected case studies, On site exploration, mapping of existing practices and generative codesign, Definition of priority design areas where innovation activities can be activated
2. Envisioning and Prototyping RQ2: How can agroecological farms operate as nodes within these territorial systems? Tasks: Development of applied concepts and preliminary prototypes, Testing of possible integrations between educational, productive, ecological and community functions, Development of scenarios describing possible evolution into regenerative rural hubs
3. Governance and replication RQ3: What governance and economic models are required to sustain these systems? Tasks: Analysis of governance arrangements and organizational models, Exploration of economic models supporting rural living systems, Identification of impact dimensions and monitoring indicators

Expected Results

The project will generate analytical, design and strategic outputs. First, the research will produce a systemic understanding of the territorial living system of Ca' Mariuccia. This phase will identify key spatial, social and technological components through the creation of an ecosystem map (D1.1), together with priority design areas (D1.2) where innovation activities can be activated. Second, the project will develop a prototype of an agroecological farm as a territorial infrastructure node (D2.1), together with scenarios (D2.2) exploring how productive, ecological and community functions can be integrated to generate rural proximity. Finally, the research will define governance and economic models and impact indicators (D3.1) and operational guidelines for replication (D3.2), complemented by policy recommendations (D3.3) supporting the development of similar territorial living systems in other small and remote places. These results will provide practical tools for farms, local organisations and public institutions engaged in territorial innovation.

Innovation and Multidisciplinarity

The project is innovative in that it explores how agroecological farms can evolve into regenerative rural hubs integrating productive, ecological and community functions within territorial living systems. While many initiatives address agriculture, energy or rural development separately, this research investigates how these dimensions can be connected through design to create integrated infrastructures supporting ecological regeneration, local economies and new forms of community in small and remote territories. The project requires a multidisciplinary approach combining design, environmental studies, territorial planning, and engineering. Design supports system mapping, prototyping, and scenario development. Environmental and energy perspectives contribute to understanding agroecological practices and resource cycles, while territorial planning and management perspectives help analyse spatial dynamics, governance conditions, and economic models. Together, these perspectives address rural territories as complex socio-ecological systems where spatial, environmental and social dimensions must be considered together to enable feasible innovation pathways.

External Institutions and Tutors

Institutions: CA' MARIUCCIA S.S.A, Azienda Agricola, Albugnano, Asti

Principal tutors: Andrea Pirollo, andrea.pirollo@camariuccia.it, founder and owner of Ca' Mariuccia

Additional tutors: Liat Rogel, liat.rogel@gmail.com, expert on collaborative housing and social innovation

Modello di rigenerazione dei suoli urbani e peri-urbani

Contact Information

Project type: Industry & Innovation oriented

Main contact: Margherita Caggiano. Re Soil Foundation. Corso Duca degli Abruzzi 24, 10129, Torino

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Overall Project Objectives

The project aims at developing a model applicable to urban and periurban areas that starting from soil regeneration can generate positive impacts on local areas. By adopting a multidisciplinary approach, which will involve best practice of soil regeneration, urban architecture, financial sustainability, and social sciences, the project will develop a model able to regenerate urban and peri-urban sites affected by soil degradation. The project will involve stakeholders of the quadruple helix and explore points of interests that can encourage local authorities, industries, academia and the civil society to actively participate in the regeneration process. To do so, the project will explore compensation model that can offer sustainability credits and other benefits to the participating stakeholders. The project will be beneficial for local areas, communities, authorities, industries, farmers, professionals, R&D actors. Ultimately, the project will define measurable KPIs and impacts and develop a protocol that can facilitate the adoption and replication of the model developed.

General Project Description

According to the EUSO, approximately 65% of soils in the EU is affected by degradation. Urban soils are highly threatened by soil sealing, compaction, contamination and loss of green spaces. The project will identify a pilot site, potentially in Turin area, that is affected by soil degradation and study potential solutions for soil

regeneration. To do so, the involvement of local stakeholders such as local authorities, industries, communities, R&D, farmers, professionals will be crucial. The project will also identify potential benefits and sustainability compensation that can encourage the participation of the private sector in order to ensure financial viability. Other funding opportunities, such as crowd-funding, funded projects may also be explored. The pilot site will become a place for social inclusiveness, in which events, training sessions, capacity building, arts and cultural initiatives will be organised to increase knowledge and engagement about the need to regenerate soils. The project will also look at ways to involve vulnerable beneficiaries. It will be crucial to identify KPIs to measure the impacts generated by the project. The project results will be used to develop a protocol to help replication. Key phases: map pre-existing experiences of urban soil regeneration; identify the characteristics of a suitable pilot site; identify the site; develop a business plan; define KPIs; identify relevant local stakeholders and reach out to them; design and test regeneration practices; identify sustainability compensation models and funding opportunities; engage stakeholders and local community; measure KPIs; develop and disseminate the protocol.

Expected Results

Develop and test a model able to regenerate soils in urban and periurban sites; map and analyse stakeholders; define governance models; identify options for financial sustainability; map beneficiaries and target audiences; identify engagement opportunities; measure impacts; develop a protocol; disseminate project results.

Innovation and Multidisciplinarity

Existing initiatives for the regeneration of urban and periurban soils are often limited to specific aspects and/or actions while this project aims to develop a holistic approach that can involve multiple disciplines, including: soil science, landscape architecture, financial models, LCAs, social science and inclusiveness, stakeholder engagement, education, arts&culture, communication and dissemination.

External Institutions and Tutors

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